BASE PLATE DESIGN CRITERIA

PURPOSE

This document establishes guidelines and recommended procedures for the design of column base plates and includes a series of base plate capacity curves for preliminary base plate selection.

SCOPE

This document includes the following major sections:

- DEFINITIONS
- BASIC THEORY
- HOLLOW STRUCTURAL SECTIONS
- FULL COMPRESSION
- PARTIAL COMPRESSION
- FULL TENSION
- CORNER BENDING
- SIDE BENDING
- BASE PLATE CAPACITY CURVES
- BIAXIAL BENDING
- BASE PLATES WITH HIGH SHEAR
- WELD DESIGN
- GROUT
- PC PROGRAMS
- REFERENCES
- ATTACHMENTS

This document covers the basic theory of base plate allowable stress design (ASD) and includes three sample calculations and a set of base plate capacity curves. This document does not cover load and resistance factor design (LRFD). Also included are short discussions and alternative base plate designs for high shear loads, base plates with biaxial bending, and general applications for grout and welds.

APPLICATION

This document applies to the design of base plates that have vertical loads (tension or compression), a moment or both and will be limited in general to base plates supporting wide flange or hollow structural section (HSS) columns. The wide flange details will be limited in general to those shown in Structural Engineering Practices:

000.215.4050 Standard Anchor Bolts and Sleeves - Details
000.215.5030 Structural Steel General Notes

DEFINITIONS

A_p BD - Area of steel base plate

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A_C	Area of the concrete pier
A_L	Area of load
A_t	Net tensile stress area of anchor bolts
a	Point of fixity for bending cantilever of base plate on compression side
B	Width of steel base plate (parallel to column flanges) – See Figure 1
b	Distance from centerline of column to centerline of anchor bolt (parallel to column web)
b_f	Width of compression flange of steel column
C	Resultant compressive force
c	Effective width for side or corner bending
D	Depth of steel base plate (parallel to column web) – See Figure 1
d	Depth of steel column
d_b	Anchor bolt diameter
d_m	Distance from centerline of bolt to edge of base plate (parallel to column web)
d_n	Distance from centerline of bolt to edge of base plate (parallel to column flanges)
e	M/P - Eccentricity of applied loads
E_C	Modulus of elasticity of concrete
E_S	Modulus of elasticity of steel
F_a	Allowable compression stress in concrete
f_a	Actual compression stress in concrete at point a
F_b, f_b	Allowable, actual bending stress in base plate
F_p, f_p	Allowable, actual bearing stress on concrete
F_t, f_t	Allowable, actual tensile stress in anchor bolts
f_c	Maximum compression stress (bearing) on concrete
f'_c	Specified compression strength of the concrete

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g	D - dm - Distance from face of column to center of tension bolts
h	Height of concrete pier
M	Applied Moment
M_{PL}	Actual moment in the steel base plate
M_{xx}, M_{yy}	Applied moments about the xx and yy axis of the column at bottom of base plate (biaxial loading)
m	(D - 0.95d)/2 - Cantilever length for bending of steel base plate (parallel to column web)
N	Ratio of modulus of elasticity of steel to concrete (E_S/E_C)
n	(B - 0.80 b_f)/2 - Cantilever length for bending of steel base plate (parallel to column flanges)
P	Applied axial load
P_T	Allowable tension in anchor bolts
S_{xx}	$BD^2/6$ - Section modulus of base plate about xx axis
S_{yy}	$DB^2/6$ - Section modulus of base plate about yy axis
T	Actual tension load in anchor bolts
t	Thickness of steel base plate
X	Distance from centerline of anchor bolts in tension to critical section in bending
Y	Distance from edge of base plate to limit of compression block
θ	Angle associated with corner bending of base plate

BASIC THEORY

The major purpose of a column base plate is to distribute the compressive load of the column over a larger area that supports the column. The base plate is sized on the

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assumption that the portion of the base plate that overhangs the column flange acts as a cantilever beam with its fixed end located just inside the column edges.

Refer to Figure 1.

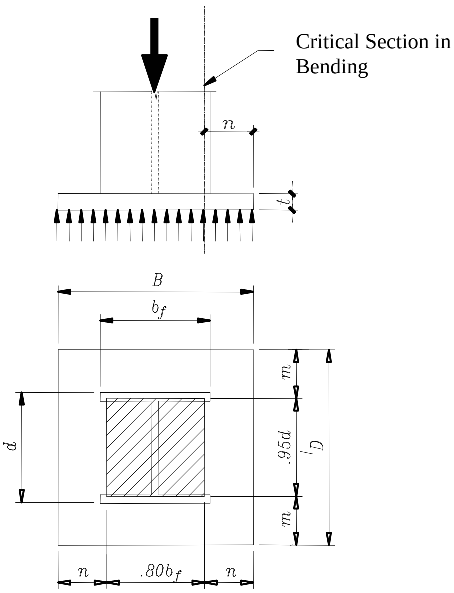

Figure 1. Base Plate Details

For base plates with very small overhangs or no overhang at all, refer to AISC (American Institute for Steel Construction) Steel Construction Manual, 13th Edition, listed in the References section.

For initial selection of base plate dimensions including anchor bolt hole distances and clearances, please refer to Structural Engineering Standard Drawing 000.215.5040: Structural Steel Connection Details.

BASIC THEORY: HOLLOW STRUCTURAL SECTIONS

Base plates for rectangular and circular pipe columns can be designed using the same procedures described for wide flange sections. The point of fixity for the assumed cantilever should be based on 0.95 times the outside column dimensions for rectangular sections and 0.80 times the outside dimension for circular sections. These values are approximations which correspond to the values used for wide flange sections. However, the area enclosed by the hollow sections is somewhat stiffer than that between the flanges of a wide flange column. Therefore, the use of these same values is conservative.

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BASIC THEORY:
FULL COMPRESSION

In most situations, typical engineered base plates can be divided into two main cases. In the first case, the base plate has a vertical compression load with or without a small moment where there is compression over the entire area between the bottom of the base plate and the pier. Refer to Figure 2. In this case, the eccentricity e is less than or equal to D/6.

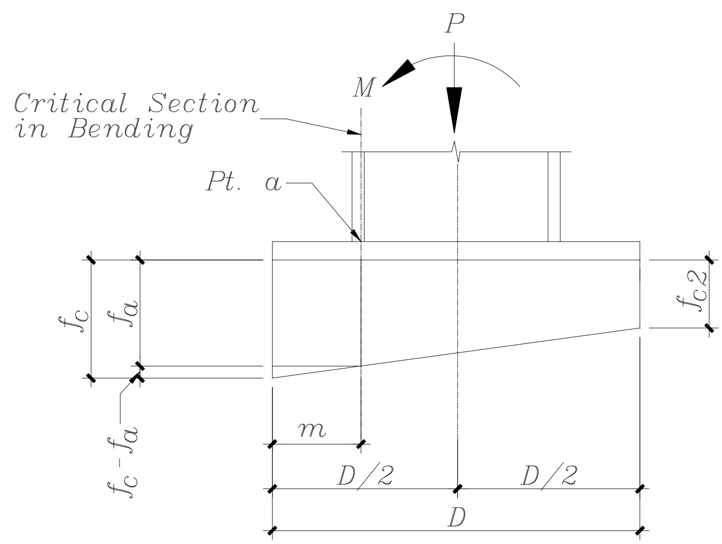

Figure 2. Base Plate with Full Compression

The bending moment in the base plate is given by the following formula:

$$M_{PL} = \frac{m^2 B}{6}\left(f_a + 2f_c\right)$$

Where $\mathrm{f_c} = \frac{1}{\mathrm{A_p}}\left[\mathrm{P} + \frac{6\mathrm{M}}{\mathrm{D}}\right]$

And $\mathrm{f_a} = \frac{1}{\mathrm{A_P}}\left[\mathrm{P} + \frac{6\mathrm{M}(\mathrm{D} - 2\mathrm{m})}{\mathrm{D}^2}\right]$

The thickness of the base plate may then be computed by the formula:

$$t \geq \sqrt{\frac{6M_{PL}}{B\,F_b}}$$

where B is the base plate width and F_b is the allowable bending stress of the plate.

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BASIC THEORY:
PARTIAL COMPRESSION

In the second case, the base plate has a vertical load and a moment which places the base plate under partial compression, increasing the compression on one side of the plate and allowing tension in the anchor bolts on the opposite side. Refer to Figure 3. In this case, the eccentricity e is greater than D/6.

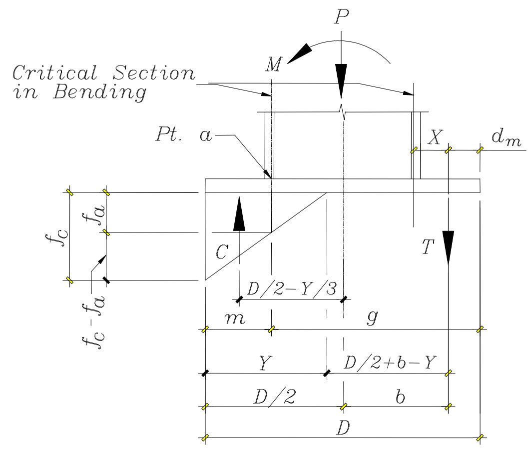

Figure 3. Base Plate with Partial Compression

The distance Y of the compression zone is calculated by the cubic equation:

$$Y^3 + K_1 Y^2 + K_2 Y + K_3 = 0$$

Where

$$K_1 = 3\left(e - \frac{D}{2}\right)$$

$$K_2 = \frac{6\,N\,A_t}{B}(b + e)$$

$$K_3 = -K_2\left(\frac{D}{2} + b\right)$$

The total tension in the anchor bolts is calculated by the formula:

$$T = -P\left[\frac{\frac{D}{2} - \frac{Y}{3} - e}{\frac{D}{2} - \frac{Y}{3} + b}\right]$$

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Anchor bolt tension should be checked. For complete anchor bolt design requirements, refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide.

The maximum compression between the base plate and the pier is then:

$$f_c = \frac{2(P+T)}{Y\,B}, \quad \text{and}$$

$$f_a = f_c\left[\frac{Y-m}{Y}\right]$$

Bearing on the concrete pier should be checked.

The maximum bending moment in the base plate will be the larger value caused by either compression under the base plate or tension from the anchor bolts. The effective length (X) for the tension case may be taken as the distance from the centerline of the anchor bolt to the critical section in bending.

$$M_{PL} = \frac{m^2 B}{6}\left(f_a + 2f_c\right) \quad \text{Compression under base plate}$$

or $$M_{PL} = T\,X \quad \text{Tension from bolts}$$

and finally, the plate thickness is, $t \geq \sqrt{\frac{6M_{PL}}{B\,F_b}}$

In the case of bending controlled by bolt tension, an AISC guide uses a smaller effective width, instead of B. However, it is felt that the entire plate width is appropriate because that is the limit state of the failure mechanism. The entire plate would need to yield before the plate would fail in bending.

For situations where the plate is much wider than the column, it is prudent to limit the effective plate width to the column width plus 2 times the distance of X, i.e. $B \le b_f + 2X$.

In some cases side or corner bending may govern the design of the base plate. Therefore, the required plate thickness for these conditions should also be checked.

For a thorough discussion of base plate formulae, refer to both Blodgett and Lother, in the References section. The accuracy of the preceding design procedure is dependent on two additional constraints: First, the anchor bolt embedment must be sufficient to develop the appropriate tensile forces, refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide. Second, the base plate must be sufficiently thick such that bolt prying forces when calculated in accordance with AISC are not significant. Normally, for Fluor Standard Details, bolt prying is not an issue. However, when using high strength anchor bolts or when bolt tension approaches the tensile capacity of the bolts the prying effects may become significant.

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BASIC THEORY:
FULL TENSION

Occasionally a third case occurs where the base plate is subject to full uplift and no portion of the base plate is under compression. In this case, the tension in the bolts can be calculated from the following formula:

$$T = P \pm \frac{M}{2b}$$

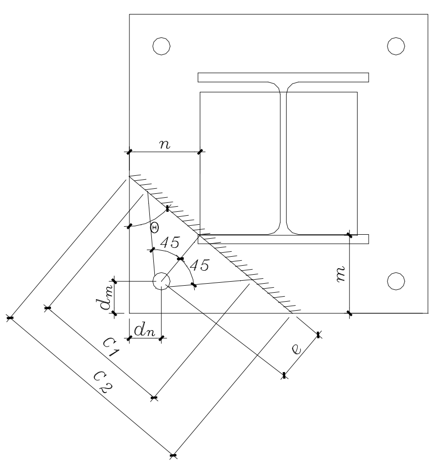
where b equals the distance from the centerline of the column to centerline of the bolt.

BASIC THEORY:
CORNER BENDING

Corner bending of the base plate should be checked when the anchor bolts are beyond the column edge. Tension on the bolts causes bending at an angle across the plate.

It is assumed that the plane of bending is at a right angle to the line from the effective column area to the bolt.

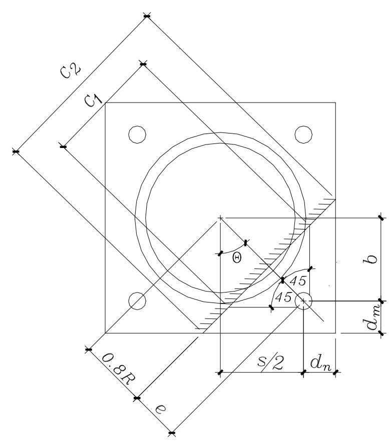
For wide flange members



$$\tan\theta = \frac{m - d_m}{n - d_n}$$

$$e = \sqrt{(m - d_m)^2 + (n - d_n)^2}$$

For circular pipe members



$$\tan\theta = \frac{\frac{s}{2} + d_n}{b + d_m}$$

$$e = \sqrt{b^2 + \left(\frac{s}{2}\right)^2} - 0.8R$$

Use the lesser of the two effective bending widths:

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$$c_1 = 2e + 1.6d_b$$

or

$$c_2 = \sqrt{(n + m\tan\theta)^2 + \left(m + \frac{n}{\tan\theta}\right)^2}$$

$$c_1 = 2e + 1.6d_b$$

or

$$c_2 = e\tan\theta + \frac{e}{\tan\theta} + \frac{d_m}{\cos\theta} + \frac{d_n}{\sin\theta}$$

Moment in base plate where T is maximum tension per anchor bolt

$$M_{PL} = Te$$

and finally, the plate thickness is : $t \geq \sqrt{\frac{6M_{PL}}{cF_b}}$

BASIC THEORY:
SIDE BENDING

Side bending of the base plate can govern when n > m.

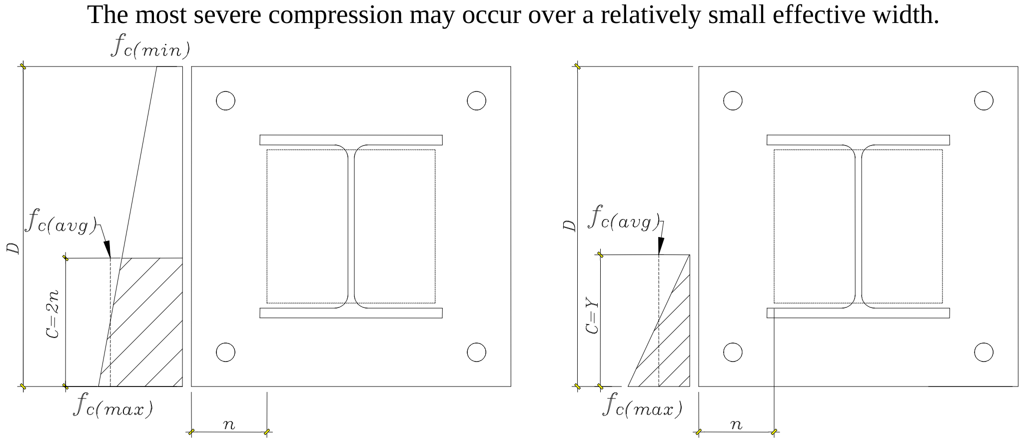
When $e \leq \frac{D}{6}$

$$f_{c(max)} = \frac{1}{A_p}\left(P \pm \frac{6M}{D}\right)$$

When $e > \frac{D}{6}$

$$f_{c(max)} = \frac{2(P+T)}{YB}$$

The most severe compression may occur over a relatively small effective width.



Therefore, examine the bending stress using an effective width, c, equal to the lesser of the following values:

$c_1 = 2n$ or $c_2 = Y$ or $c_3 = D$

The maximum value of 2n for the effective width was chosen as a reasonable approximation for the average peak stress in the corner arm. If the dimension Y is very small, a minimum value for the effective width of c = n is recommended.

Take the average compressive stress over the effective bending width.

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$$f_{c(avg)} = \frac{f_{c(\max)} + f_{c(c)}}{2}$$

Find the moment in the plate due to side bending.

$$M_{PL} = \frac{f_{c(avg)}}{2}(c)n^2$$

Finally, the required base plate thickness is found.

$$t \geq \sqrt{\frac{6M_{PL}}{0.75(c)F_Y}}$$ thickness required due to bending

$$t \geq \frac{f_{c(avg)}\, n}{0.4F_Y}$$ thickness required due to shear

BASE PLATE CAPACITY CURVES

Attachment 05 contains a set of base plate capacity (load-versus-moment) curves for various anchor bolt diameters and base plate thicknesses for the standard base plate sizes shown in Structural Engineering Standard Drawing 000.215.5030: Structural Steel General Notes. These curves were generated by back calculating the formulae from the Basic Theory section for the maximum P and M depending on whether the base plate is under complete compression, partial compression or tension, or complete tension. The derivation of these formulae can be found in Shipp. Refer to the References section.

Because several assumptions are required to develop the P versus M curves (such as anchor bolt capacity, loaded concrete area), the curves should be used only for initial base plate and anchor bolt sizing, and then verified either by hand calculation or computer analysis.

All charts are based on the following assumptions:

- Moments are in one direction only and are about the strong axis of the column.
- Base plate material is ASTM A36. In case ASTM A572 Gr.50 base plate is being used multiply the thickness by a factor of $\sqrt{36ksi / 50ksi} = 0.8485$
- The square root of the ratio of the concrete area to the base plate area equals 1.3:
  $$\sqrt{\frac{A_c}{A_L}} = 1.3$$
  This represents a typical value for a steel pipe support type column on a concrete pier.
- Anchor bolt capacities are based on 20 ksi (138 MPa) allowable stress on the net tensile area minus 1/8 of an inch (3 mm) for corrosion. Refer to Attachment 04. Shear loads on anchor bolts have not been considered. For complete anchor bolt design requirements, refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide.
- Anchor bolts are located a nominal 1-1/2 inches (38 mm) from the flange face of the

BASE PLATE DESIGN CRITERIA

column in the strong axis of the column.

To use the base plate capacity curves, enter the charts with the applied moment and vertical load and select the plate thickness and anchor bolt size from the first curve to the right.

Note!!! Divide M and P by 1.33 for load combination with wind or seismic as permitted by the code used.

BIAXIAL BENDING

In most cases, base plates are analyzed for moment in one direction only. In the event of significant moments acting concurrently in two directions such as in a corner column, a special analysis is required.

Typically, there are two basic cases; one is full compression under the base plate and the other is partial compression under the base plate and tension in one or more anchor bolts.

In the case of full compression under the base plate, the stress at any point is the algebraic sum of the stresses due to the axial load and the moments about the two principal axes XX and YY.

The maximum compression at the corner of the base plate is then the following:

$$f_c = \left(\frac{P}{A}\right) \pm \left(\frac{M_{xx}}{S_{xx}}\right) \pm \left(\frac{M_{YY}}{S_{YY}}\right)$$

A sample calculation is included in Attachment 03 (Attachment 03M).

In the case where there is only partial compression under the base plate and tension in one or more anchor bolts, there is no easy closed form solution. The problem is most easily solved by performing trial and error in locating the neutral axis. For a complete discussion of this topic, refer to MacGinley, in the References section.

BASE PLATES WITH HIGH SHEAR

In most cases, the base plate design is not governed by vertical shear on the plate. However, this failure mechanism should be investigated for very thick plates subject to high axial loads.

One method to avoid a very thick base plate is to add stiffeners or anchor bolt chairs to distribute the load on the base plate. These design procedures are not covered in this guideline.

Base plates are normally not affected by lateral shear loads; however, high shear loads can be critical for anchor bolts. For the design of anchor bolts, refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide. Two possibilities may be considered for reducing high shear loads in the anchor bolts. One

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alternative is to sandblast the bottom of the base plate and coat it with an epoxy primer. After the column and base plate have been installed and leveled, nonshrink epoxy grout is poured between the base plate and pier. Shear loads are then transferred directly from the base plate to the pier through bond of the epoxy grout. Bond values between the steel and the epoxy grout are required from the grout manufacturer. This procedure requires careful construction techniques and should be considered as a less preferred method.

The other alternative to accommodate high lateral shear loads is to provide shear plates or a shear key on the bottom of the base plate, grouted into preformed slots in the concrete pier. In this case, shear loads are transferred from the base plate to the pier by direct bearing of the shear plates against the grout and concrete. For the proper embedment, it is important that grout completely fills the slots in the pier. Therefore, the use of grout holes in the base plate is recommended. Additional calculations are required to ensure that the allowable bearing and shear stresses are not exceeded in the pier. For shear key design, please refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide.

When shear is directly transferred from the base plate to the anchor bolts, bearing strength at bolt holes shall be checked per the AISC specification.

WELD DESIGN

When designing the welds, only the weld on either side of the web should be considered to transfer the shear force parallel to the web from the column into the baseplate. When designing flange welds for tension, the welds on both sides of the near flange can be considered.

GROUT

There are two primary types of grout used underneath base plates. The most widely used are poured-in-place proprietary non-shrink grout mixes. This grout is purchased premixed, and water is added until the mix is flowable. Small forms are placed on the pier, and the grout mix is poured from one end thus forcing the air out the other side. Purchased grouts are more expensive than drypack; however, the savings in reduced labor costs and the reduction in installation time commonly outweigh the initial material costs.

The other type of grout is drypack. Drypack is mixed as required by using sand and cement with just enough water to form a stiff paste. This paste is then placed under the base plate and rodded with wood posts to provide continuous bearing between the base plate and the pier.

PC PROGRAMS

The types of calculations and iterations required for base plate design are ideally suited to a computer solution. These calculations can be performed by the base plate program(s) available on the Fluor Network.

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REFERENCES

AISC (American Institute of Steel Construction), Steel Construction Manual, 13th Ed., 2005

AISC (American Institute of Steel Construction), Hollow Structural Sections Connection Manual (pg 7-9, 7-10), 1st Ed., 1997

AISC (American Institute of Steel Construction), Steel Design Guide Series, Column Base Plates, 1990

ASCE (American Society of Civil Engineers), Wind Loads and Anchor Bolt Design for Petrochemical Facilities, 1st Ed., New York, 1997

ACI (American Concrete Institute), Building Code Requirements for Structural Concrete, ACI 318-05/ACI 318M-05

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Blodgett, Omer W., Design of Welded Structures, The James F. Lincoln Arc Welding Foundation, Cleveland, June 1966

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MacGinley, T. J., Structural Steelwork Calculations and Detailing, Butterworths. London, 1973

Shipp, J. G., Design of Base Plates for Axial and Moment Loads, Technical Paper Number 628. Fluor Engineering Library, Irvine, CA. August 1978

Structural Engineering
Guideline 000.215.STE05121: Anchor Bolt Design Guide

Structural Engineering
Standard 000.215.4050: Standard Anchor Bolts and Sleeves - Design Details

Structural Engineering
Standard 000.215.5030: Structural Steel General Notes

Structural Engineering
Standard 000.215.5040: Structural Steel Connection Details

ATTACHMENTS

Attachment 01: **(31Mar05)**
Sample Design 1 (US Units): Base Plate Design - Full Compression

Attachment 01M: **(26Jun2007)**

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Sample Design 1 (SI Units): Base Plate Design - Full Compression

Attachment 02: (31Mar05)
Sample Design 2 (US Units): Base Plate Design - Partial Compression

Attachment 02M: (26Jun2007)
Sample Design 2 (SI Units): Base Plate Design - Partial Compression

Attachment 03: (31Mar05)
Sample Design 3 (US Units): Base Plate Design - Bi-Axial Loads

Attachment 03M: (26Jun2007)
Sample Design 3 (SI Units): Base Plate Design - Bi-Axial Loads

Attachment 04: (31Mar05)
Allowable Anchor Bolt Tension Loads Used for Base Plate Capacity Curves (US Units)

Attachment 05: (31Mar05)
Base Plate Capacity Curves (US Units)

Note!!! Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.

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DESIGN EXAMPLE 1 – BASE PLATE – Full Compression

Design Data

Concrete - f'_c = 3000 psi

Steel - ASTM A-36

Column - W12 x 40

Pier - 2′-0″ Square

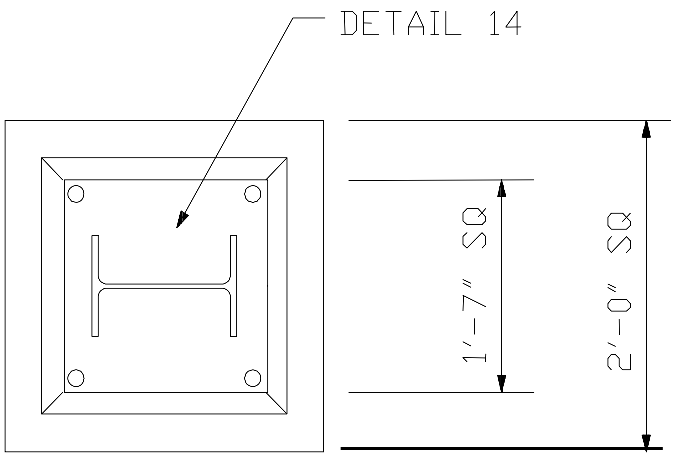
Anchor Bolts - 1″ dia. headed A.B.

(see Standard 670.215.4050)

Base Plate Detail 14

(see Standard 670.215.5030)

Codes - AISC 13th Edition (ASD)



Design Loads

P = 80K DL + LL

M = $10^{l\text{-}k}$ DL + LL

For examples of load development, see the following guidelines:

000.215.1215 -Wind Loads

000.215.1216 -Earthquake Loads

000.215.1250 -Pipe Supports

Analysis

B = 19″ D = 19″ b_f = 8″ d = 12″

m = (D - 0.95d) / 2 = (19″ - 0.95(12)) / 2 = 3.80″

n = (B - 0 .8b_f) / 2 = (19″ - 0.8(8")) / 2 = 6.30″

e = M / P = ($10^{l\text{-}k}$)(12″) / 80K = 1.5″

D / 6 = 19″ / 6 = 3.17″ > e

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DESIGN EXAMPLE 1 – BASE PLATE – Full Compression

$$f_c = \frac{1}{A_p}\left[P + \frac{6M}{D}\right] = \frac{1}{19^2}\left[80K + \frac{6\left(10'^{-K}\right)(12)}{19''}\right] = 0.327\,ksi$$

$$f_a = \frac{1}{A_p}\left[P + \frac{6M(D-2m)}{D^2}\right] = \frac{1}{19^2}\left[80K + \frac{6(10)(12)(19-2(3.8))}{19^2}\right] = 0.285\,ksi$$

$$f_{c_{ALLOW}} = 0.35 f'_c \sqrt{A_C / A_L} = (0.35)(3\ KSI)\sqrt{24^2/19^2} = 1.33\,ksi \;>\; 0.327\,ksi \qquad OK \;\left(\text{Re}\,f\text{: } AISC\ J9\right)$$

$$M_{PL} = \frac{m^2 B}{6}\left(f_a + 2f_c\right) = \frac{(3.8)^2(19'')}{6}\left(0.235\,ksi + 2(0.327\,ksi)\right) = 42.94''^{-K}$$

Check Side Bending:

Full compression ==> Check side bending over lesser of D, or 2n.

2n = 2 (6.3″) = 12.6″

$$f_{c(\max)} = \frac{1}{19^2}\left[80 + \frac{6(120)}{19}\right] = 0.327\ ksi$$

$$f_{a(\max)} = \frac{1}{19^2}\left[80 + \frac{6(120)\left(19-(2)2n\right)}{19^2}\right] = 0.187\ ksi$$

$$f_{c(avg)} = \frac{f_{c(\max)} + f_{a(\max)}}{2} = 0.257\ ksi$$

$$M_{PL} = \frac{n^2\,2n\,f_{c(avg)}}{2} = \frac{(6.3)^2(12.6'')(0.257\text{ ksi})}{2} = 64.3''^{-K} \quad > \quad 42.94''^{-K}$$

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DESIGN EXAMPLE 1 – BASE PLATE – Full Compression

$$t = \sqrt{\frac{6M_{PL}}{2nF_b}} \qquad F_b = (0.75)(36\ ksi) = 27\ ksi \qquad (\text{Ref}: AISC\ F2)$$

$$\text{t} = \sqrt{\frac{6(64.3)}{(12.6)(27)}} = 1.23"$$

USE 1.25” PLATE

BASE PLATE DESIGN CRITERIA

Sample Design 1M (ASD - SI Units): Base Plate -- Full Compression

Note: Not all calculations in this sample are updated per latest codes.

Design Data

Concrete - f'_c = 21 MPa

Steel - ASTM A-36, f_y = 250 MPa

Column – W310 x 60

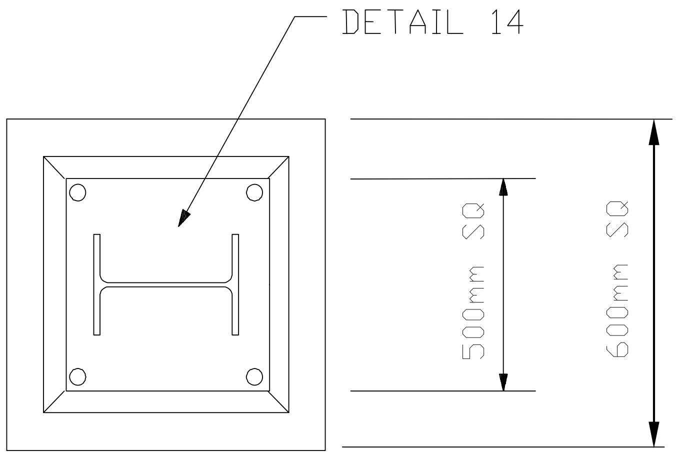
Pier – 600 mm Square

Anchor Bolts – M27 headed A.B.

(see Standard Dwg. 000.215.4050)

Base Plate Detail 14

(see Standard Dwg. 000.215.5030)



Codes - AISC 13th Edition (ASD)

Design Loads

P = 360 kN, DL + LL

M = 14 kN-m, DL + LL

For examples of load development, see the following guidelines:

000.215.1215 -Wind Loads

000.215.1216 -Earthquake Loads

000.215.1250 -Pipe Supports

Analysis

B = 500 mm D = 500 mm b_f = 203 mm d = 303 mm

m = (D - 0.95d) / 2 = (500mm - 0.95(303mm)) / 2 = 106 mm

n = (B - 0 .8b $_f$) / 2 = (500mm - 0.8(203mm)) / 2 = 169 mm

Check Strong Axis Bending:

e = M / P = (14kN-m)(1000) / 360kN = 39 mm

BASE PLATE DESIGN CRITERIA

Sample Design 1M (ASD - SI Units): Base Plate -- Full Compression

D / 6 = 500mm / 6 = 83 mm > e

$$f_c = \frac{1}{A_p}\left[P + \frac{6M}{D}\right] = \frac{1}{(500mm)^2}\left[360kN + \frac{6(14kN\text{-}m)}{500mm}\right] = 2.11MPa$$

$$f_a = \frac{1}{A_p}\left[P + \frac{6M(D-2m)}{D^2}\right] = \frac{1}{(500mm)^2}\left[360kN + \frac{6(14kN\text{-}m)(1000)(500mm - 2(106mm))}{(500mm)^2}\right] = 1.83MPa$$

$$f_{c_{ALLOW}} = 0.35 f'_c \sqrt{A_C / A_L} = (0.35)(21MPa)\sqrt{(600mm)^2/(500mm)^2} = 8.8MPa$$

$$> 2.11MPa \quad OK \quad (Ref: AISC \;\; J9)$$

$$M_{PL} = \frac{m^2 B}{6}(f_a + 2f_c) = \frac{(106mm)^2(500mm)}{6}(1.83MPa + 2(2.11MPa)) = 5.67kN\text{-}m$$

$$F_b = 0.75Fy = (0.75)(250\ MPa) = 188MPa \qquad (Ref: AISC\ F2)$$

$$t = \sqrt{\frac{6M_{PL}}{B F_b}} = \sqrt{\frac{6(5.67kN\text{-}m)}{(500mm)(188MPa)}} = 19mm$$

Check Base Plate Flexural Yielding inside the Column Flange:

$$n' = \frac{\sqrt{d\, b_f}}{4} = \frac{\sqrt{303mm(203mm)}}{4} = 62mm \; < m = 106mm$$

Yielding inside the column flange will not control. See AISC ASD pg 3-106, Column Base Plates, Design procedures.

Check Shear under Strong Axis Bending:

$$V = \frac{f_c + f_a}{2} mB = \frac{2.11MPa + 1.83MPa}{2}(106mm)(500mm) = 104.5kN$$

$$F_v = 0.4Fy = (0.4)(250\ MPa) = 100MPa \qquad (Ref: AISC\ ASD\ Spec.\ Sect.\ F4)$$

$$t = \frac{V}{F_v B} = \frac{104.5kN}{(100MPa)(500mm)} = 2.1mm$$

BASE PLATE DESIGN CRITERIA

Sample Design 1M (ASD - SI Units): Base Plate -- Full Compression

Check Side Bending:

Full compression ==> Check side bending over lesser of D, or 2n.

2n = 2 (169mm) = 338 mm < D

$$f_{c(\max)} = \frac{1}{A_p}\left[P + \frac{6M}{D}\right] = \frac{1}{(500mm)^2}\left[360kN + \frac{6(14kN\text{-}m)(1000)}{500mm}\right] = 2.11MPa$$

$$f_{a(\max)} = \frac{1}{A_p}\left[P + \frac{6M(D-2n)}{D^2}\right] = \frac{1}{(500mm)^2}\left[360kN + \frac{6(14kN\text{-}m)(1000)(500mm - 2(338mm))}{(500mm)^2}\right]$$
$$= 1.2MPa$$

$$f_{c(avg)} = \frac{f_{c(\max)} + f_{a(\max)}}{2} = 1.66MPa$$

$$M_{PL} = \frac{n^2 (2n)\, f_{c(avg)}}{2} = \frac{(169mm)^2 (338mm)(1.66MPa)}{2} = 8.0kN\text{-}m \quad > 5.67kN\text{-}m$$

$$F_b = 0.75Fy = (0.75)(250\ MPa) = 188MPa \qquad (Ref\ :\ AISC\ F2)$$

$$t = \sqrt{\frac{6M_{PL}}{(2n)F_b}} = \sqrt{\frac{6(8kN\text{-}m)}{(338mm)(188MPa)}} = 27.5mm\ \ (controls)$$

Check Shear under Side Bending:

By comparison, shear load along the side bending line is slightly larger than that in the case of strong axis bending but the magnitude does not govern the design.

SUMMARY: USE 30mm PLATE

Bearing Strength at Bolt Holes:

No lateral shear transfer through anchor bolts ==> no need to check.

Weld Design:

Full compression, no lateral shear ==> Use minimum welds. Minimum fillet weld size is 8 mm for plate thickness > 19 mm (Ref AISC ASD Table J2.4). Also refer to Structural Engineering Standard Drawing: 000.215.5040 Structural Steel Connection Details.

USE 10 mm weld at flanges, both sides and 8 mm weld at web, both sides.

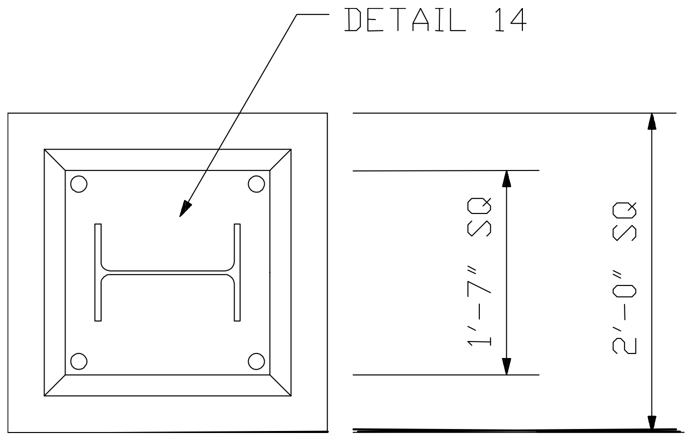
BASE PLATE DESIGN CRITERIA

Sample Design 1M (ASD - SI Units): Base Plate -- Full Compression

BASE PLATE DESIGN CRITERIA

Sample Design 2: Baseplate Design -- Partial Compression

Design Data



Concrete - f'_c = 4000 PSI

Steel - ASTM A-36

Column - W12 x 40

Pier - 2′-0″ Square

Anchor Bolts - 1″ dia. headed A.B.

(See Practice 000.215.4050)

BasePlate Detail 14

(See Practice 000.215.5030)

Codes - AISC 13th Edition, ACI 318-05

Design Loads

P = 40K DL + LL

M = 40 $^{l\text{-}k}$ Wind

For examples of load development see guidelines:

000-215-1215 - Wind Loads

000-215-1216 - Earthquake Loads

000-215-1250 - Pipe Supports

Analysis

B = 19″ D = 19″ b_f = 8″ d = 12″

m = (D - 0.95d) / 2 = (19″ - 0.95(12″)) / 2 = 3.80″

n = (B - 0.8b_f) / 2 = [19″ - 0.8(8″)] / 2 = 6.30″

e = M / P = (40$^{\prime\text{-}K}$)(12″) / 40 K = 12″

D / 6 = 19 / 6 = 3.17″ < 12″

BASE PLATE DESIGN CRITERIA

Sample Design 2: Baseplate Design -- Partial Compression

$$Y^3 + K_1Y^2 + K_2Y + K_3 = 0$$

$$K_1 = 3\left(e - \frac{D}{2}\right) = 3\left(12'' - \frac{19''}{2}\right) = 7.5$$

$$K_2 = \frac{6NA_T}{B}(b+e)$$

$$N = \frac{E_s}{E_c} = \frac{29000\ ksi}{57\sqrt{4000}} = 8.04 \qquad \text{(Ref : AISC 6 - 3 and ACI 8.5)}$$

$$A_T = (2)(.606\ in^2) = 1.21\ in^2 \qquad \text{(Ref : Practice 000.215.1207)}$$

$$b = d/2 + X = 12''/2 + 1.5 = 7.5''$$

$$K_2 = \frac{(6)(8.04)(1.21)}{19}(7.5+12) = 59.91$$

$$K_3 = -K_2(D/2 + b) = -59.91(19/2 + 7.5) = -1018.47$$

$$Y^3 + 7.5Y^2 + 59.91Y - 1018.47 = 0$$
$$Y = 6.63$$

$$T = -P\left[\frac{\frac{D}{2} - \frac{Y}{3} - e}{\frac{D}{2} - \frac{Y}{3} + b}\right] = -40\left[\frac{\left(\frac{19}{2}\right) - \left(\frac{6.63}{3}\right) - 12}{\left(\frac{19}{2}\right) - \left(\frac{6.63}{3}\right) + 7.5}\right] = 12.74K$$

$$T_{ALLOW} = (2)(8.9K)(1.33) = 23.67K > 12.74K \qquad OK$$

Note!!! For detailed instructions on anchor bolt design, see Guideline 000.215.1207, Anchor Bolt Design Criteria.

$$f_c = \frac{2(P+T)}{YB} = \frac{2(40K + 12.74K)}{(6.63'')(19'')} = 0.837\ ksi$$

$$f_{c(ALLOW)} = 0.35 f'_c\sqrt{A_C/A_L} \qquad (\text{Ref: } AISC\ J9)$$

BASE PLATE DESIGN CRITERIA

Sample Design 2: Baseplate Design -- Partial Compression

$$\sqrt{A_C / A_L} = \sqrt{\frac{(24)^2}{(6.63)(19)}} = 2.14$$

$\boxed{\text{USE } 2.0}$

$$f_{c(ALLOW)} = 0.35(4)(2.0)(1.33) = 3.724\ ksi \quad >> 0.837\ ksi$$

$$f_a = f_c\left[\frac{Y-m}{Y}\right] = 0.837\left[\frac{6.63-3.8}{6.63}\right] = 0.357\ ksi$$

$$M_{PL} = \frac{m^2 B}{6}(f_a + 2 f_c) = \frac{(3.8)^2(19)}{6}(0.357 + 2(0.837)) = 92.88''^{-K} \qquad (Compression\ Side) \quad \leftarrow CONTROLS$$

$$M_{PL} = T\,X = (12.74K)(1.5'') = 19.11''^{-K} \qquad (Tension\ Side)$$

Check Side Bending:

Partial compression ==> Check side bending over lesser of Y or 2n.

Y = 6.63″ < 2n = 12.6″

$$f_{c(avg)} = \frac{40 kips}{(19 in)6.63 in} = 0.317 \text{ ksi}$$

$$\mathrm{M_{PL}} = \frac{n^2 Y\ f_{c(avg)}}{2} = \frac{(6.3)^2(6.63)(0.836)}{2} = 55.1''^{-K}$$

$$F_b = (.75)(36)(1.33) = 36\ KSI \qquad (\text{Re}\,f{:}\ \ AISC\ F2)$$

$$t = \sqrt{\frac{6M_{PL}}{DF_b}} = \sqrt{\frac{6(92.88)}{19(36)}} = 0.903'' \qquad (\text{for normal bending})$$

$$t = \sqrt{\frac{6M_{PL}}{YF_b}} = \sqrt{\frac{6(55.1)}{6.63(36)}} = 1.18'' \qquad (\text{for side bending})$$

BASE PLATE DESIGN CRITERIA

Sample Design 2: Baseplate Design -- Partial Compression

USE 1.25″ PLATE ← Controls

Check Corner Bending:

$$\tan^{-1}\left(\frac{m-dm}{n-dn}\right)=\theta \qquad \tan^{-1}\left(\frac{3.8"-2"}{6.3"-2"}\right)=22.7^\circ$$

$$\mathrm{e}=\sqrt{(m-dm)^2+(n-dn)^2} \qquad \mathrm{e}=\sqrt{(3.8"-2")^2+(6.3"-2")^2}=4.66$$

$$c_1=2e+1.6d_b=10.3"$$

$$c_2=\sqrt{(n+m\tan\theta)^2+\left(m+\frac{n}{\tan\theta}\right)^2}=20.2"$$

$$M_{PL}=\frac{Te}{2}=\frac{12.74kip(4.66")}{2}=29.8k"$$

$$t_{PL}=\sqrt{\frac{6M_{PL}}{c_1F_b}}=\sqrt{\frac{6(29.8k")}{10.3"(36ksi)}}=0.674"$$

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

Note: Not all calculations in this sample are updated per latest codes.

Design Data

Concrete - f'_c = 28 MPa

Steel - ASTM A-36, f_y = 250 MPa

Column – W310x60

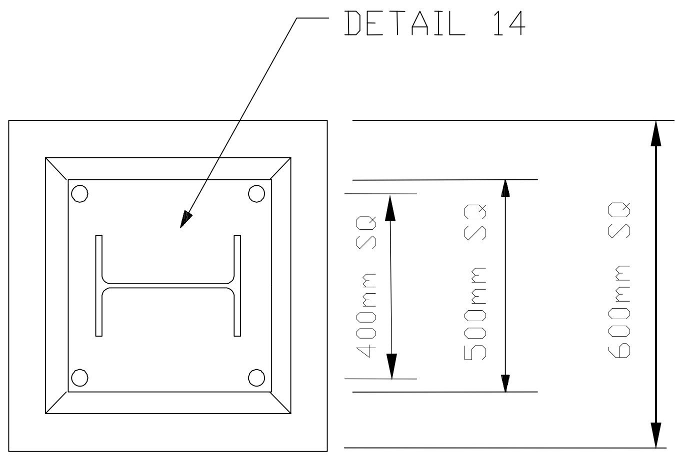
Pier – 600 mm Square

Anchor Bolts – M24 headed A.B.

(See Standard Dwg. 000.215.4050)

Base Plate Detail 14

(See Standard Dwg. 000.215.5030)



Codes - AISC 13th Edition (ASD), ACI 318M-05

Design Loads

P = 180 kN, DL + LL

M = 54 kN-m, Wind

For examples of load development see guidelines:

000.215.1215 - Wind Loads

000.215.1216 - Earthquake Loads

000.215.1250 - Pipe Supports

Analysis

B = 500 mm   D = 500 mm

B_{pier} = 600 mm D_{pier} = 600 mm

B_{bolt} = 400 mm D_{bolt} = 400 mm

b_f = 203 mm   d = 303 mm

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

m = (D - 0.95d) / 2 = (500mm - 0.95(303mm)) / 2 = 106 mm

n = (B - 0 .8b $_f$) / 2 = (500mm - 0.8(203mm)) / 2 = 169 mm

Check Strong Axis Bending:

e = M / P = (54kN-m)/(1000) / 180kN = 300 mm

D / 6 = 500mm / 6 = 83 mm < e

$Y^3 + K_1Y^2 + K_2Y + K_3 = 0$

$$K_1 = 3\left(e - \frac{D}{2}\right) = 3\left(300mm - \frac{500mm}{2}\right) = 0.15m$$

$$N = \frac{E_s}{E_c} = \frac{200000MPa}{4700\sqrt{28}} = 8.04 \qquad \text{(Ref : AISC 6-3 and ACI 318M-05 8.5)}$$

$$A_T = (2)(353mm^2) = 706mm^2 \qquad \text{(Ref : Guideline 000.215.1207)}$$

$$b = D_{bolt} / 2 = 400mm/2 = 200mm$$

$$K_2 = \frac{6N\,A_T}{B}(b+e) = \frac{(6)(8.04)(706)mm^2}{500mm}(200mm + 300mm) = 0.034m^2$$

$$K_3 = -K_2\left(\frac{D}{2} + b\right) = -0.034m^2\left(\frac{500mm}{2} + 200mm\right) = -0.015m^3$$

$$Y^3 + 0.15Y^2 + 0.034Y - 0.015 = 0$$
$$Y = 0.172m \text{ (or } 172\,mm)$$

$$T = -P\left[\frac{\frac{D}{2} - \frac{Y}{3} - e}{\frac{D}{2} - \frac{Y}{3} + b}\right] = -180kN\left[\frac{\left(\frac{500mm}{2}\right) - \left(\frac{172mm}{3}\right) - 300mm}{\left(\frac{500mm}{2}\right) - \left(\frac{172mm}{3}\right) + 200mm}\right] = 49.14kN$$

Note!!! For anchor bolt design, see Guideline 000.215.1207, Anchor Bolt Design Criteria.

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

$$f_c = \frac{2(P+T)}{Y\,B} = \frac{2(180kN + 49.14kN)}{(172mm)(500mm)} = 5.34MPa$$

$$f_{c(ALLOW)} = 0.35 f_c' \sqrt{A_C / A_L} \qquad \text{(Ref : AISC J9)}$$

$$\sqrt{A_C / A_L} = \sqrt{\frac{(600mm)(600mm - 500mm + 172mm)}{(172mm)(500mm)}} = 1.38$$

$$f_{c(ALLOW)} = 0.35(28MPa)(1.38)(1.33) = 17.96MPa > 5.34MPa$$

$$f_a = f_c\left[\frac{Y-m}{Y}\right] = 5.34MPa\left[\frac{172mm - 106mm}{172mm}\right] = 2.04MPa$$

$$M_{PL} = \frac{m^2 B}{6}(f_a + 2f_c) = \frac{(106mm)^2(500mm)}{6}(2.04 + 2(5.34))MPa$$
$$= 11.92kN\text{-}m \quad (Compression\ Side) \leftarrow CONTROLS$$

$$M_{PL} = T\,X = (49.14kN)(56.08mm) = 2.76kN\text{-}m \qquad (Tension\ Side)$$

$$F_b = (.75)(250MPa)(1.33) = 250MPa \qquad \text{(Ref : AISC F2)}$$

$$t = \sqrt{\frac{6M_{PL}}{B\,F_b}} = \sqrt{\frac{6(11.92kN\text{-}m)}{500mm(250MPa)}} = 24.0mm \qquad (for\ normal\ bending)$$

Check Base Plate Flexural Yielding inside the Column Flange:

$$n' = \frac{\sqrt{d\,b_f}}{4} = \frac{\sqrt{303mm(203mm)}}{4} = 62mm < m = 106mm$$

Yielding inside the column flange will not control. See reference AISC ASD 3-106 Column Base Plates, Design procedures.

Check Shear under Strong Axis Bending:

$$V = \frac{f_c + f_a}{2}m = \frac{5.26MPa + 2.05MPa}{2}(106mm) = 0.39\ kN/mm$$
$$F_v = 0.4Fy(1.33) = (0.4)(250\ MPa)(1.33) = 133MPa \qquad (Ref\ :\ AISC\ F4)$$
$$t = \frac{V}{F_v} = \frac{0.39kN/mm}{133MPa} = 2.9mm$$

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

Check Side Bending:

Partial compression ==> Check side bending over lesser of Y or 2n.

Y = 174mm < 2n = 338mm

$$f_{c(avg)} = \frac{f_C}{2} = \frac{5.26MPa}{2} = 2.63MPa$$

$$M_{PL} = \frac{n^2 Y \ f_{c(avg)}}{2} = \frac{(169mm)^2(174mm)(2.63MPa)}{2} = 6.5kN\text{-}m$$

$$t = \sqrt{\frac{6M_{PL}}{Y\ F_b}} = \sqrt{\frac{6(6.5kN\text{-}m\ )}{174mm(250MPa)}} = 30.2mm \qquad (for\ side\ bending) \quad \boxed{\leftarrow \text{Controls}}$$

Check Shear under Side Bending:

By comparison, shear load along the side bending line is slightly larger than that in the case of strong axis bending but the magnitude does not govern the design.

Check Corner Bending due to Anchor Bolt Tension:

$$\theta = \tan^{-1}\left(\frac{m-d_m}{n-d_n}\right) = tan^{-1}\left(\frac{106mm-50mm}{169mm-50mm}\right) = 25.27^\circ$$

$$e = \sqrt{(m-d_m)^2+(n-d_n)^2} \qquad e = \sqrt{(106mm-50mm)^2+(169mm-50mm)^2} = 131mm$$

$$c_1 = 2e+1.6d_b = 2(131mm)+1.6(24mm) = 301mm$$

$$c_2 = \sqrt{(n+m\tan\theta)^2+\left(m+\frac{n}{\tan\theta}\right)^2} = \sqrt{(169+106\tan(25.27^\circ))^2+\left(106+\frac{169}{\tan(25.27^\circ)}\right)^2} = 513mm$$

$$M_{PL} = \frac{Te}{2} = \frac{(48.65kN)(131mm)}{2} = 3.2kN\text{-}m$$

$$t = \sqrt{\frac{6M_{PL}}{c_1 F_b}} = \sqrt{\frac{6(3.2kN\text{-}m\ )}{301mm(250MPa)}} = 16.1mm \qquad (for\ corner\ bending)$$

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

Check Shear under Corner Bending:

$$V = \frac{T}{2(c_1)} = \frac{48.65kN}{2(301mm)} = 0.08kN / mm$$

By comparison, shear load does not govern.

SUMMARY: USE 30mm PLATE

Bearing Strength at Bolt Holes:

No lateral shear transfer through anchor bolts ==> no need to check.

Weld Design:

Design for tension load. Assume tension resistance provided by welds at flanges.

$$T = \frac{M}{d - t_f} - P\frac{A_{flange}}{A_W} = \frac{54kN\text{-}m}{303mm - 13mm} - 180kN\frac{2639mm^2}{7548mm^2} = 123.3kN$$

where, A_W – area of W-shape

A_{flange} – area of the flange

d - depth of W-shape

t_f – flange thickness

Available weld length, L = 2b_f = 406 mm

Tensile strength of the welds, F_{E70XX} = 485 MPa

$$F_{w(allow)} = (.3)F_{E70XX}(1.33) = (.3)(485MPa)(1.33) = 194MPa \qquad (Ref : AISC\ Table\ J2.5)$$

$$w2 = \frac{T}{F_{w(allow)}L(0.707)} = \frac{123.3kN}{194MPa(406mm)(0.707)} = 2.2mm$$

Minimum fillet weld size is 8 mm for plate thickness > 19 mm (Ref AISC ASD Table J2.4). Also refer to Structural Engineering Standard Drawing: 000.215.5040 Structural Steel Connection Details.

BASE PLATE DESIGN CRTITERIA

Sample Design 2M (ASD - SI Units): Base Plate Design -- Partial Compression

USE 10 mm weld at flanges on both sides.

Design for shear load. Assume shear resistance provided by welds at web. No shear loads in this case. Minimum weld size is 8 mm for plate thickness > 19 mm.

USE 8 mm weld at web on both sides.

BASE PLATE DESIGN CRITERIA

Sample Design 3: Baseplate Design – Bi-Axial Loading

Design Data

Concrete - f'_c = 3000 PSI

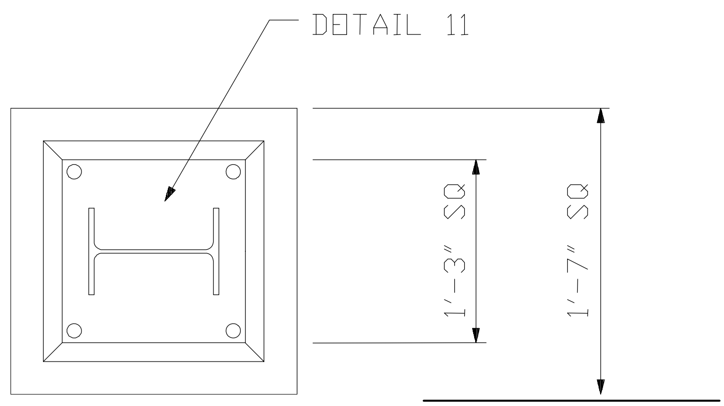
Steel - ASTM A-36

Column - W8 x 31

Pier - 1′-7″ Square

Anchor Bolts – ¾″ diam. headed A.B.
(See Standard 670.215.4050)

BasePlate - Detail 11
(See Standard 670.215.5030)



Codes - AISC 13th Edition (ASD)

Design Loads

P = 50K DL

$M_X = 7^{\text{l-k}}$ LL

$M_Y = 7^{\text{l-k}}$ LL

For examples of load development see these guidelines:

000.215.1215 - Wind Loads

000.215.1216 - Earthquake Loads

000.215.1250 - Pipe Supports

Analysis

B = 15″ D = 15″ b_f = 8″ d = 8″

$e_X = e_Y = M / P = (7^{\text{l-k}})(12^{''/'}) / 50K = 1.68''$

$D / 6 = 15'' / 6 = 2.50 > 1.68$ (Therefore Compression over entire BasePlate)

$S_X = S_Y = BD^2 / 6 = (15)(15)^2 / 6 = 562.5\ \text{in}^3$

BASE PLATE DESIGN CRITERIA

Sample Design 3: Baseplate Design – Bi-Axial Loading

$f_{b(X)}$ = M / S = ($7^{1\text{-k}}$)($12^{''/'}$) / 562.5 in^3 = .149 KSI

$f_{b(Y)}$ = M / S = ($7^{1\text{-k}}$)($12^{''/'}$) /562.5 in^3 = .149 KSI

fc = P / Ap = 50K / 152 = .222 KSI

Σ =.521 KSI (Maximum compression at corner of BasePlate)

$$f_{c(ALLOW)} = 0.35 f'_c \sqrt{A_C / A_L} = 0.35(3\,\text{ksi})\sqrt{(19)^2/(15)^2}$$
$$= 1.33\,\text{KSI} > 0.521\text{KSI} \qquad \text{(Ref : AISC J9)} \qquad \text{OK}$$

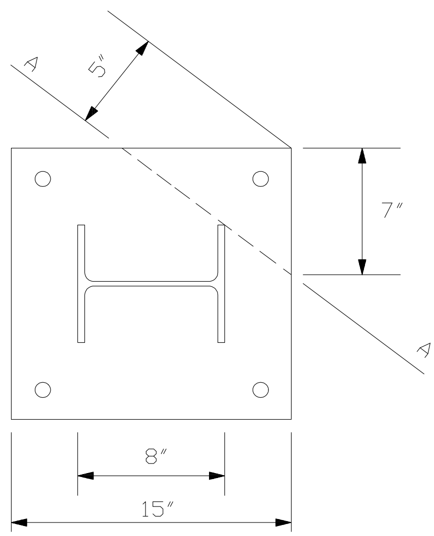
CHECK PLATE BENDING ABOUT A-A. CONSERVATIVELY USE THE MAXIMUM COMPRESSION OVER THE ENTIRE CORNER SECTION.

$$AA = \sqrt{(7)^2 + (7)^2} = 9.9''$$ **USE 10.0"**

$$M_{PL} = (\frac{1}{3})(5'')(\frac{1}{2})(7'')(7'')(0.521\,ksi) = 21.27''$$

$$F_b = (.75)(36\,ksi) = 27\,ksi$$ (Ref: AISC F2)

$$t = \sqrt{\frac{6M_{PL}}{BF_b}} = \sqrt{\frac{6\left(21.27^{''-kips}\right)}{(10'')(27\,ksi)}} = 0.688''$$



USE ¾″ PLATE

BASE PLATE DESIGN CRITERIA

Sample Design 3M (ASD - SI Units): Base Plate Design – Bi-Axial Loading

Note: Not all calculations in this sample are updated per latest codes.

Design Data

Concrete - f'_c = 21 MPa

Steel - ASTM A-36, f_y = 250 MPa

Column – W200x46

Pier – 500 mm Square

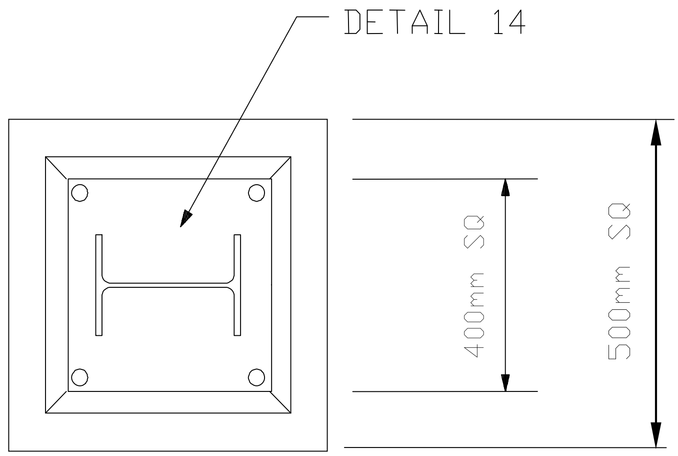
Anchor Bolts – M20 headed A.B.

(See Standard Dwg. 000.215.4050)

Base Plate Detail 14

(See Standard Dwg. 000.215.5030)

Codes - AISC 13th Edition (ASD)



Design Loads

P = 220 kN, DL

M_X = 10 kN-m, LL

M_Y = 10 kN-m, LL

For examples of load development see these guidelines:

000.215.1215 - Wind Loads

000.215.1216 - Earthquake Loads

000.215.1250 - Pipe Supports

Analysis

B = 400mm D = 400mm b_f = 203mm d = 203mm

$e_X = e_Y$ = M/P = (10kN-m)/ 220kN = 45 mm

D / 6 = 400mm / 6 = 67 mm > 45 mm (Therefore Compression over entire Base Plate)

BASE PLATE DESIGN CRITERIA

Sample Design 3M (ASD - SI Units): Base Plate Design – Bi-Axial Loading

$S_X = S_Y = BD^2 / 6 = (400mm)(400mm)^2 / 6 = 1.07x10^7\ mm^3$

$f_{b(X)} = f_{b(Y)} = M / S = (10kN\text{-}m)/(1.07x10^7\ mm^{3)} = 0.94\ MPa$

$f_c = P / A_p = 220kN /(0.16m^2) = 1.37\ MPa$

$f_{c(max)} = f_c + f_{b(X)} + f_{b(Y)} = 3.25\ MPa$ (Maximum compression at corner of base plate)

$$f_{c(ALLOW)} = 0.35 f'_c \sqrt{A_C / A_L} = 0.35(21MPa)\sqrt{(500mm)^2 / (400mm)^2}$$
$$= 9.19MPa > 3.25MPa \quad \text{(Ref : AISC J9)} \quad \text{OK}$$

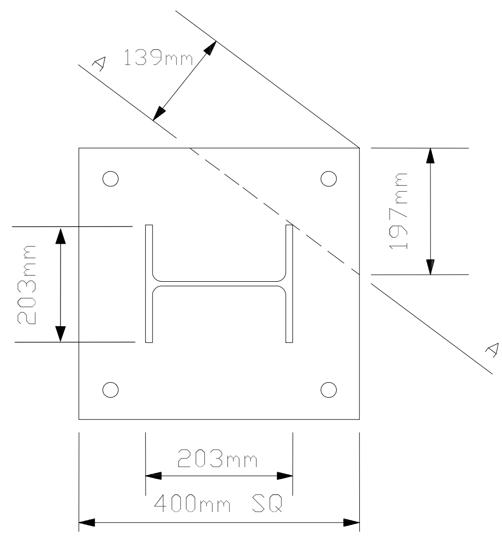
Check plate bending about A-A. Conservatively use the maximum compression over the entire corner section. From base plate and column dimensions find length of the bending line AA:

$$AA = \sqrt{(197mm)^2 + (197mm)^2} = 279mm$$

$$M_{PL} = (\frac{1}{3})(139mm)(\frac{1}{2})(279mm)(139mm)(3.25MPa)$$
$$= 2.93kN\text{-}m$$

$$F_b = (.75)(250MPa) = 187.5MPa \quad \text{(Ref: AISC F2)}$$

$$t = \sqrt{\frac{6M_{PL}}{B F_b}} = \sqrt{\frac{6(2.93kN\text{-}m)}{(279mm)(187.5MPa)}} = 18.3mm$$



BASE PLATE DESIGN CRITERIA

Sample Design 3M (ASD - SI Units): Base Plate Design – Bi-Axial Loading

Check Vertical Shear. Conservatively use the maximum compression over the entire corner section:

$$V = \frac{(139mm)(279mm)}{2}(3.25MPa) = 63kN$$

$$F_v = 0.4Fy = (0.4)(250\ MPa) = 100MPa \qquad (Ref\ :\ AISC\ F4)$$

$$t = \frac{V}{F_v(AA)} = \frac{63kN}{(100MPa)(279mm)} = 2.3mm$$

SUMMARY: BENDING CONTROLS, USE 20mm PLATE

BASE PLATE DESIGN CRITERIA

Allowable Anchor Bolt Tension Loads for Base Plate Capacity Curves

Anchor Bolt Diameter (inch)	Net Tensile Area ($inch^2$)	Allowable Tension (kips)
3/4	0.219	4.38
1.0	0.445	8.9
1- 1/4	0.763	15.26
1- 1/2	1.155	23.12

Note!!! 1. For design of anchor bolts, refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide.

2. Net Tensile Stress Area includes a corrosion allowance of 1/8″.

3. $$\text{Net Tensile Stress Area} = 0.7854\left[(\text{D} - 0.125) - \frac{0.9743}{n}\right]^2$$

Where

D = Nominal anchor bolt diameter

n = Number of thread per inch

Refer to AISC Manual of Steel Construction for more details on tensile stress area.

4. Allowable tension is based on 20 ksi allowable stress on the net tensile stress area.

5. Net tensile stress area used in the base plate capacity curves is based on a corrosion allowance of 1/8 of an inch which is typically used for nongalvanized anchor bolts. For anchor bolts that are galvanized in which no corrosion allowance is considered, the Base Plate Capacity Curves will be conservative.

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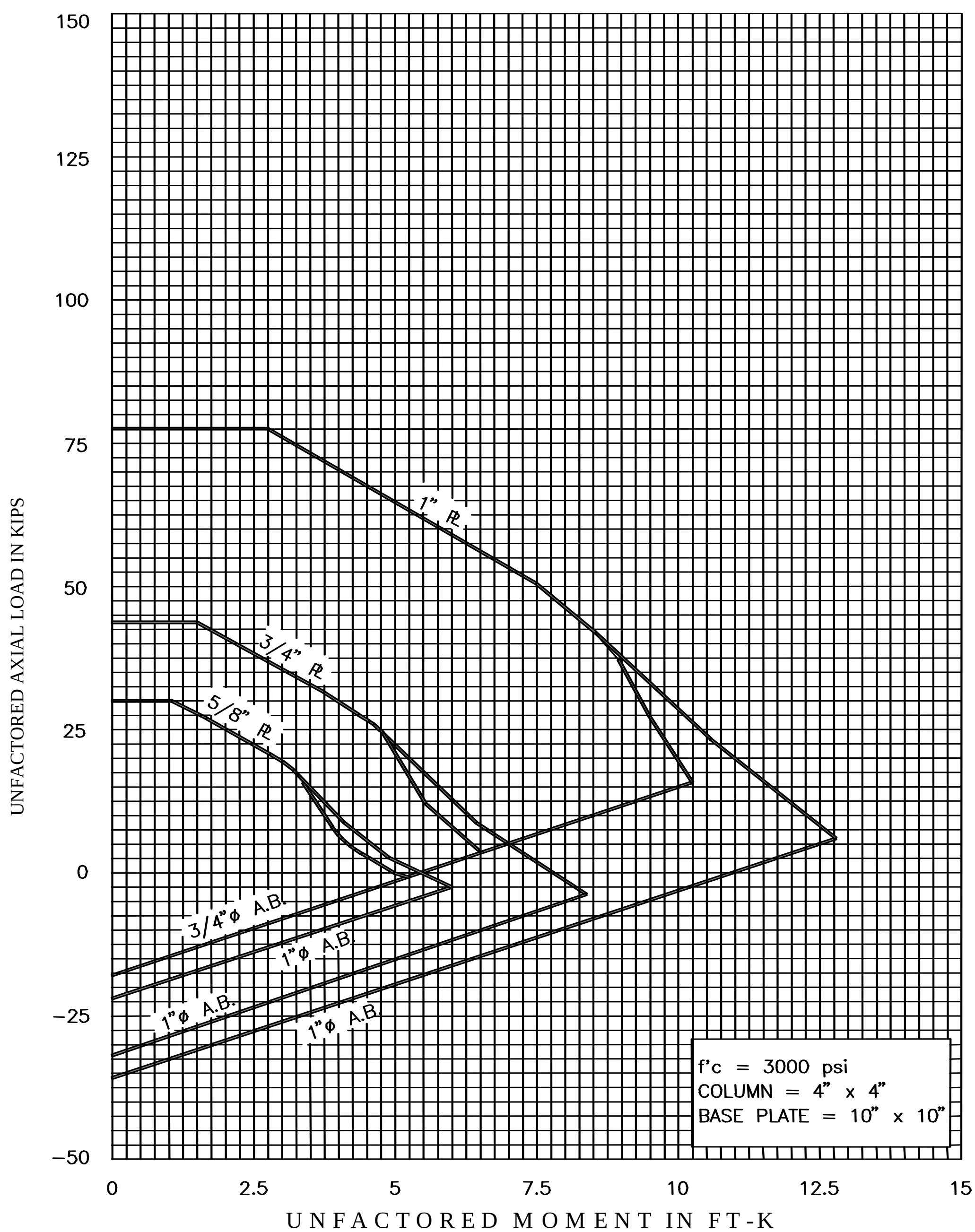
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 1

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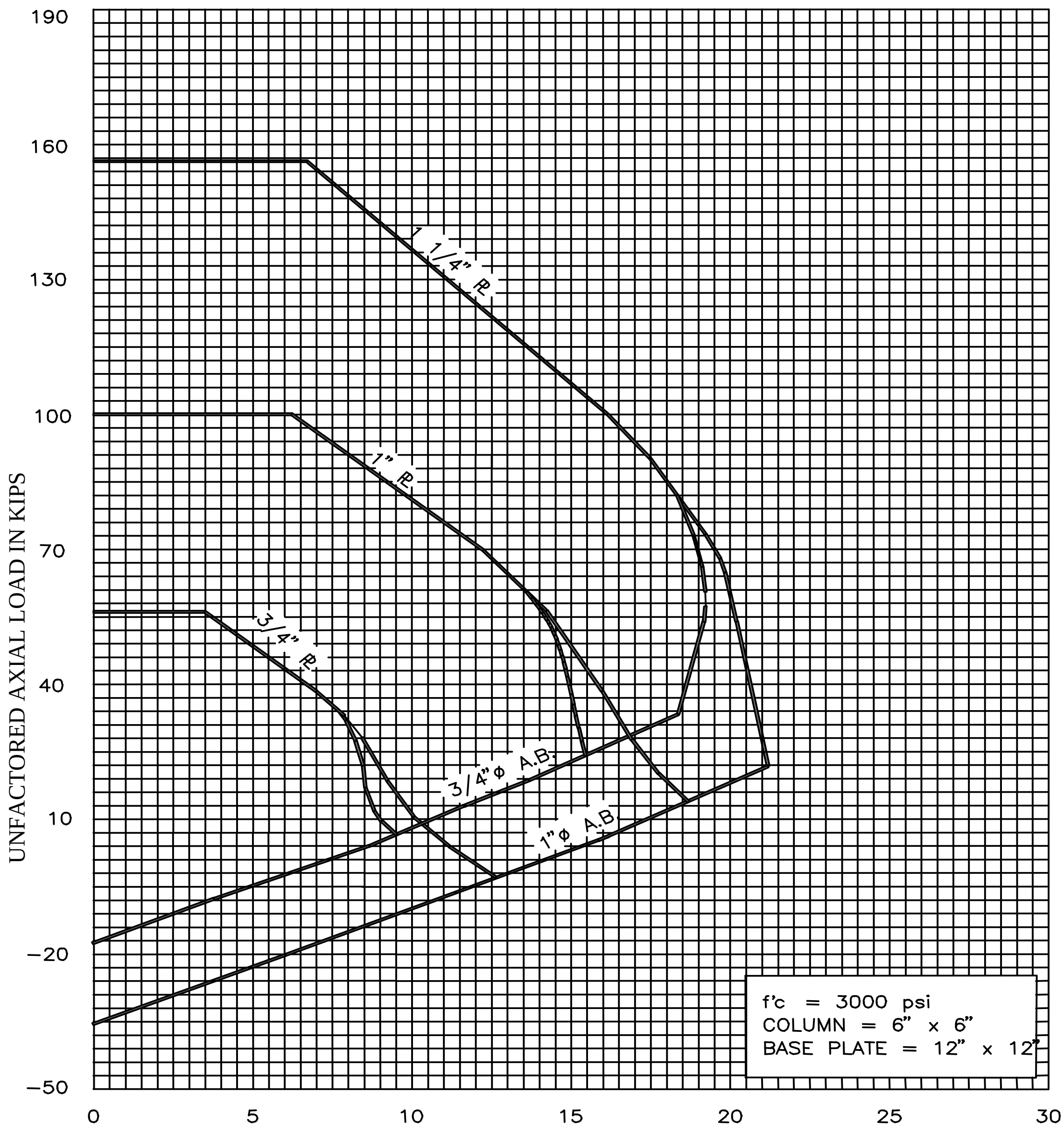
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 2

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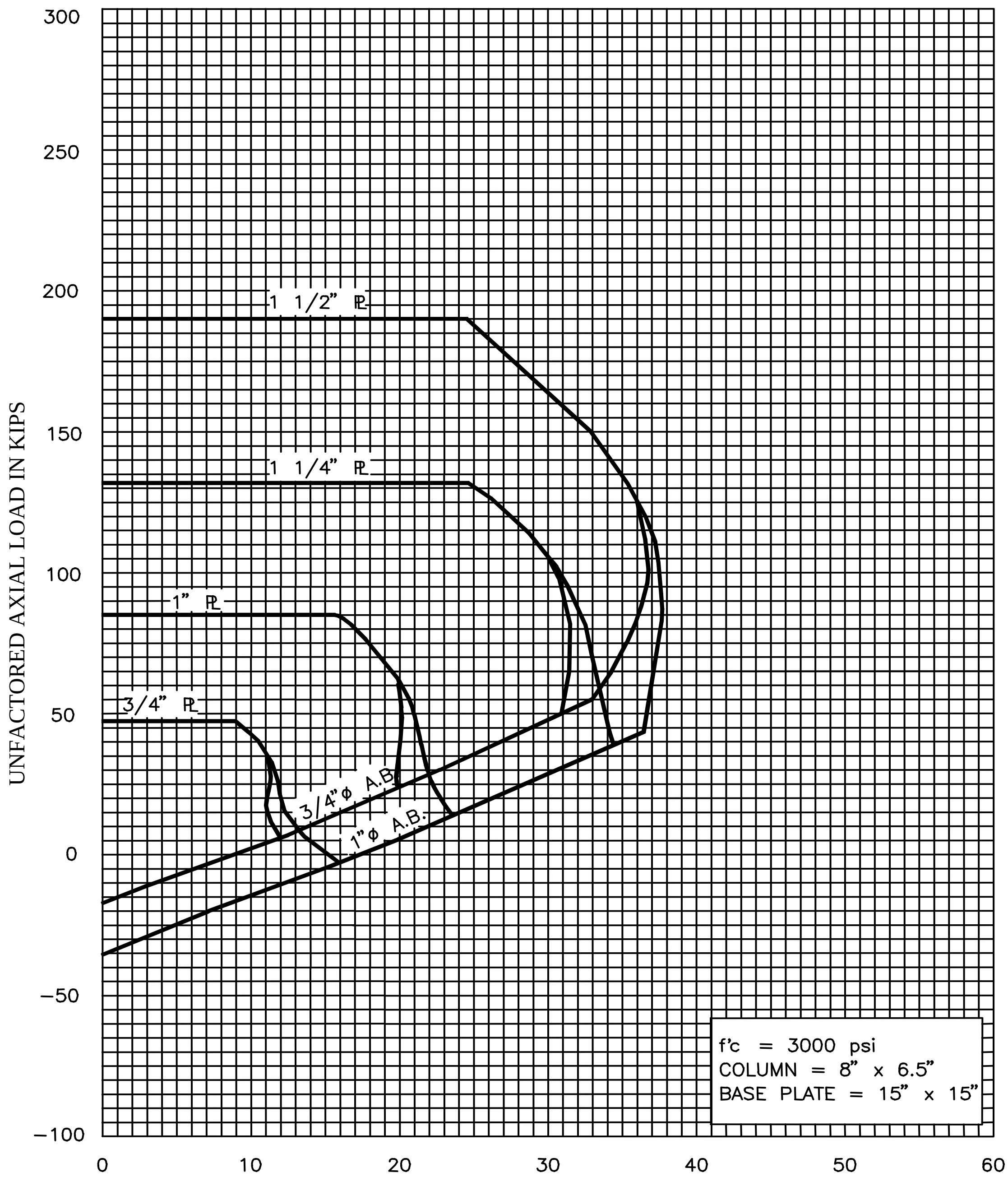
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K

Base Plate Capacity Curve 3

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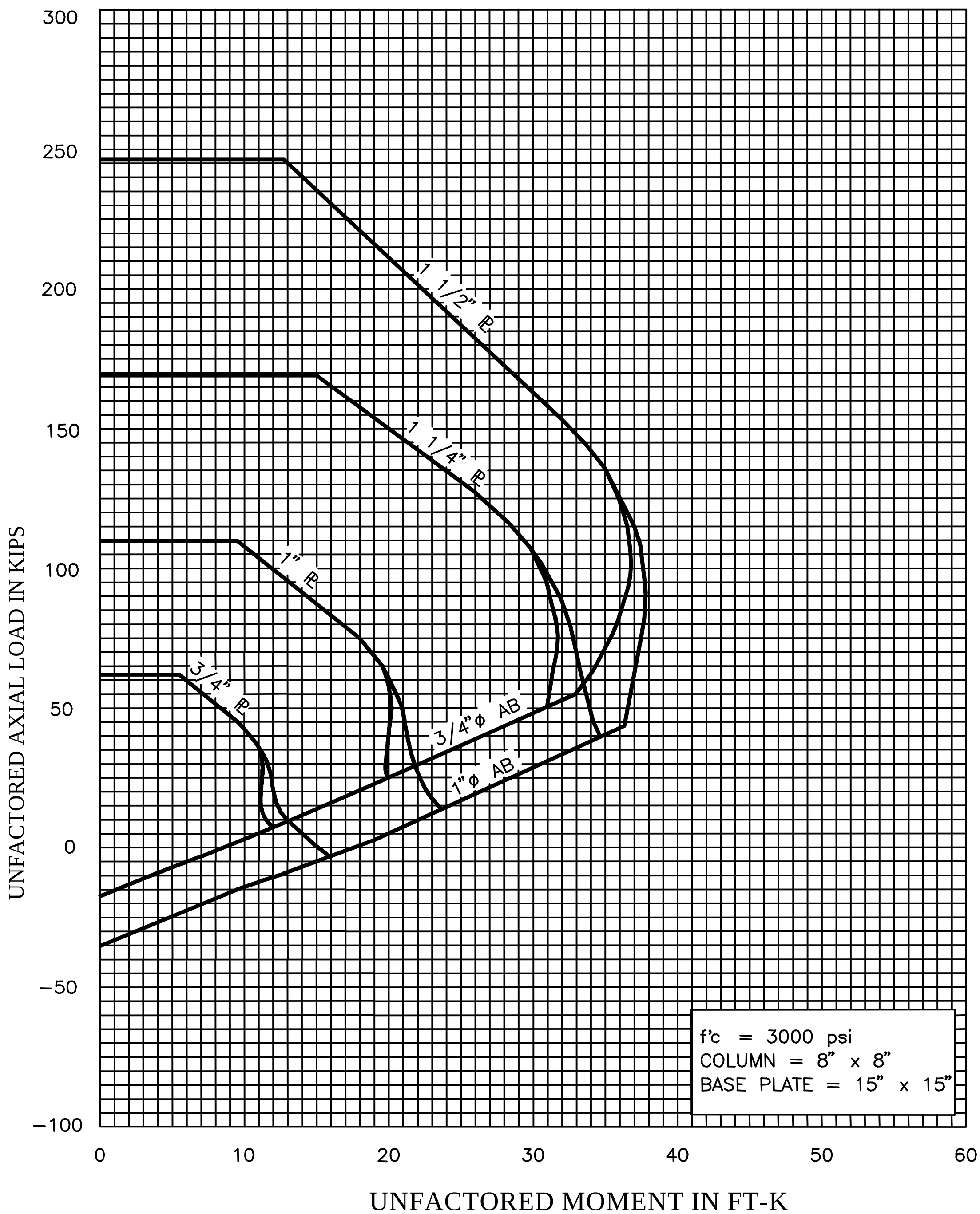
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves

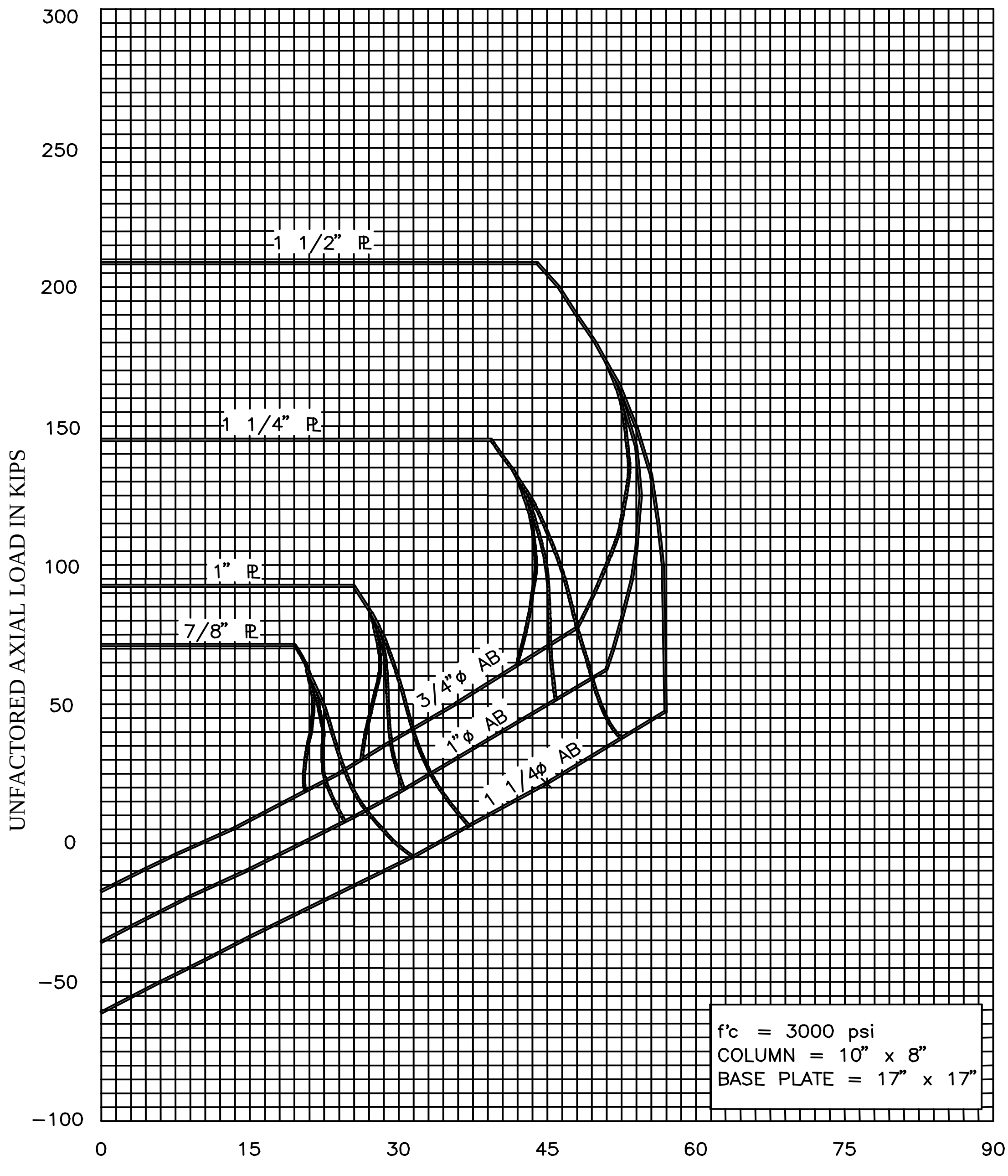


Base Plate Capacity Curve 4

FLUOR®

BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base late Capacity Curve 5

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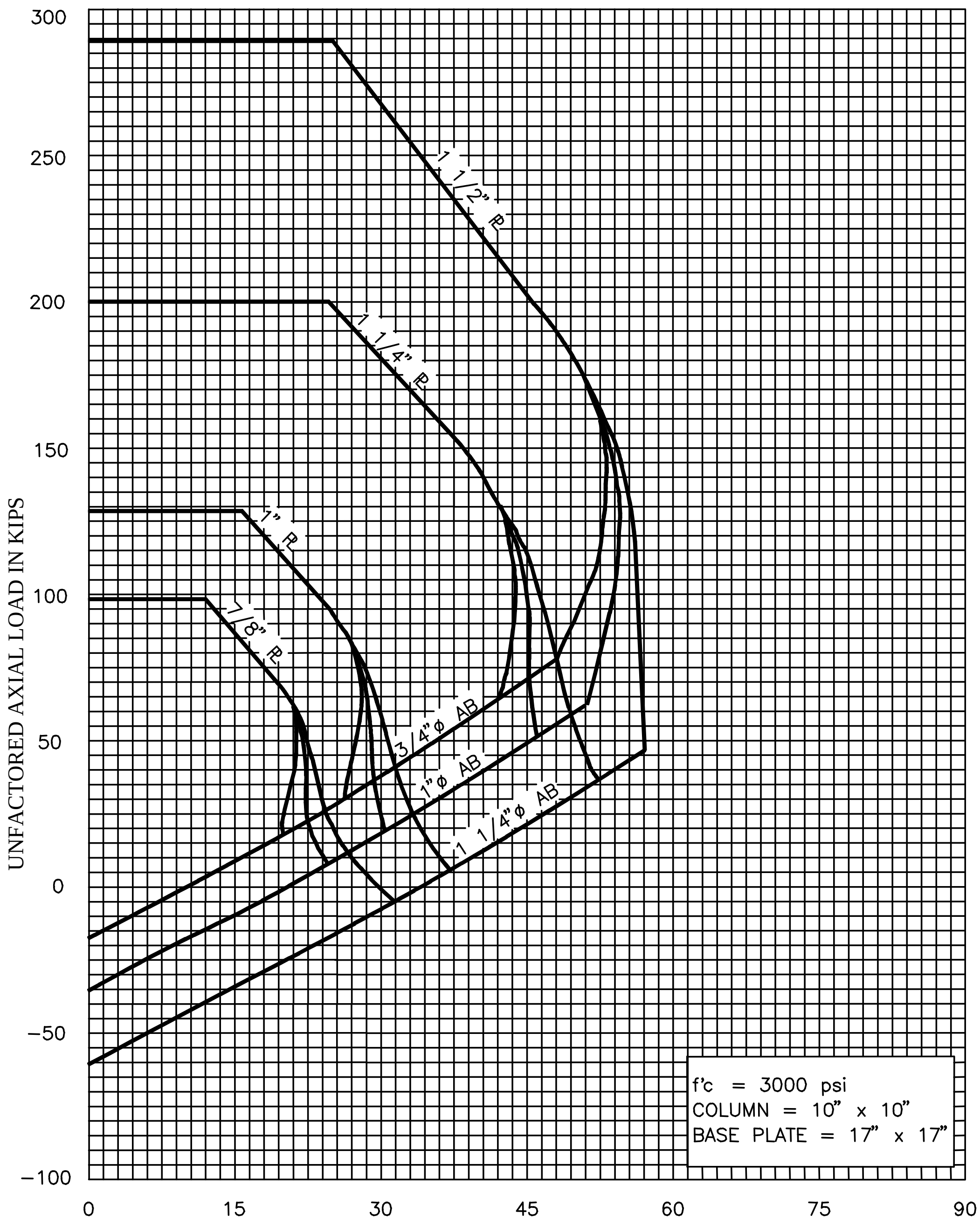
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves

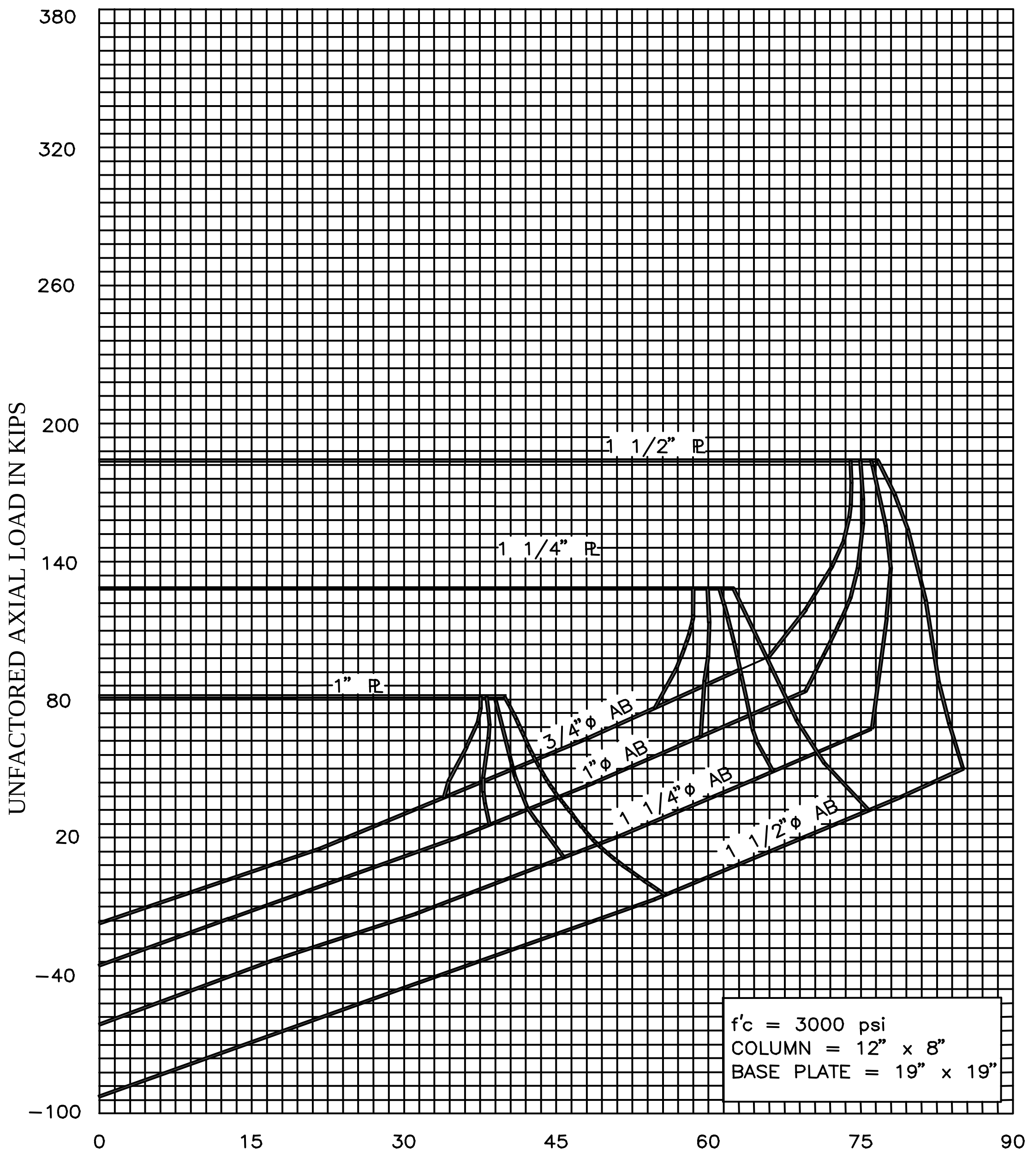


UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 6

FLUOR®

BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 7

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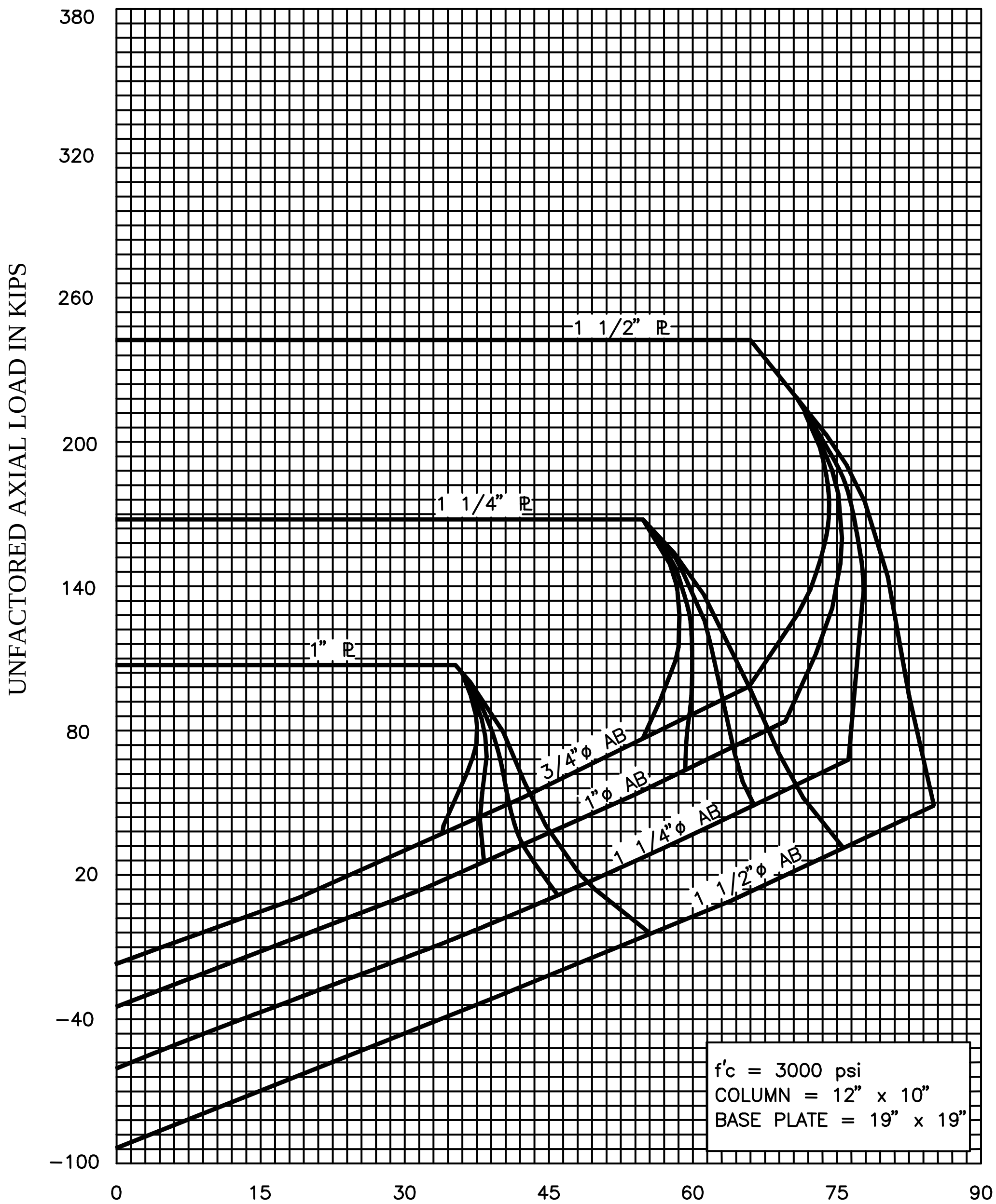
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 8

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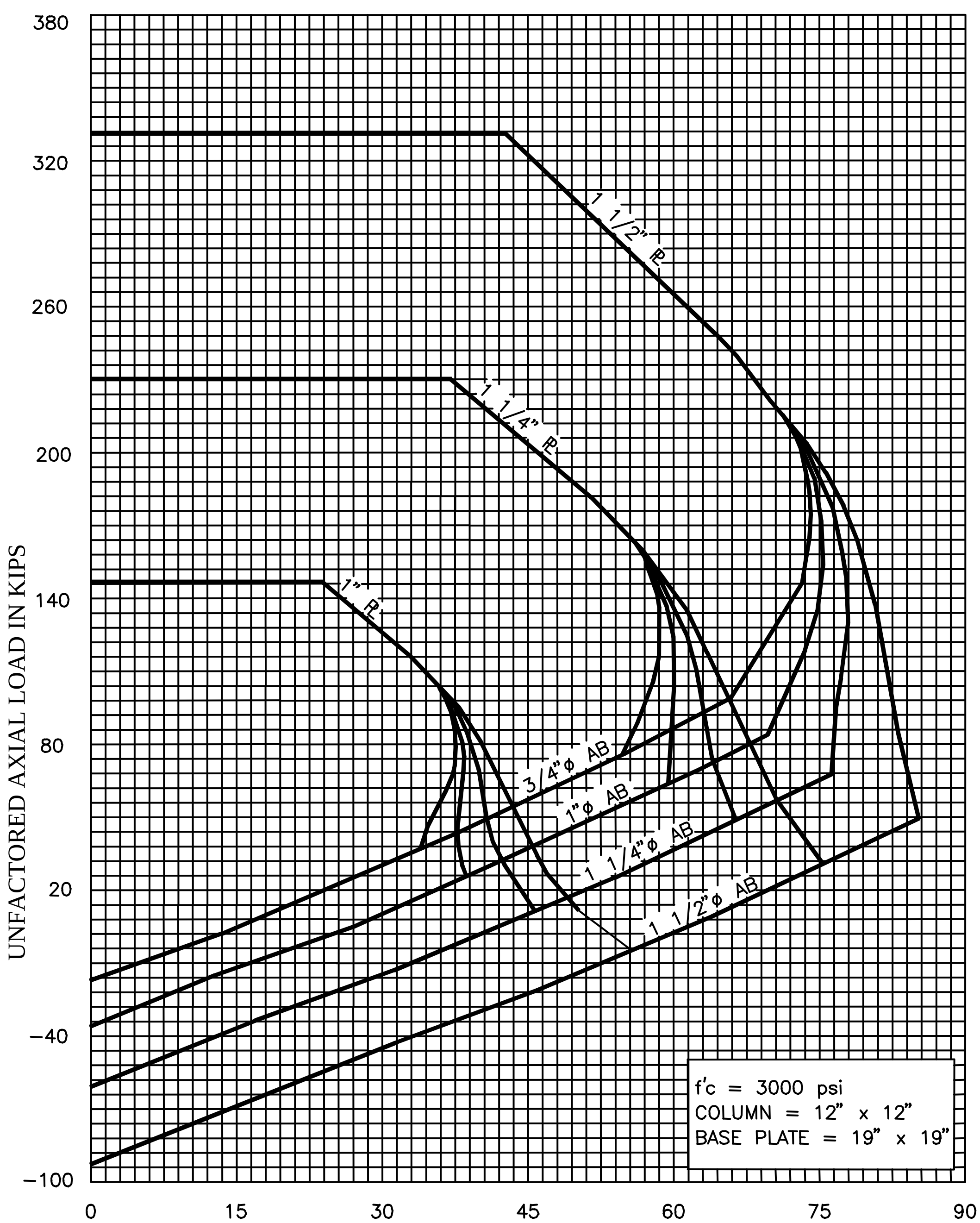
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 9

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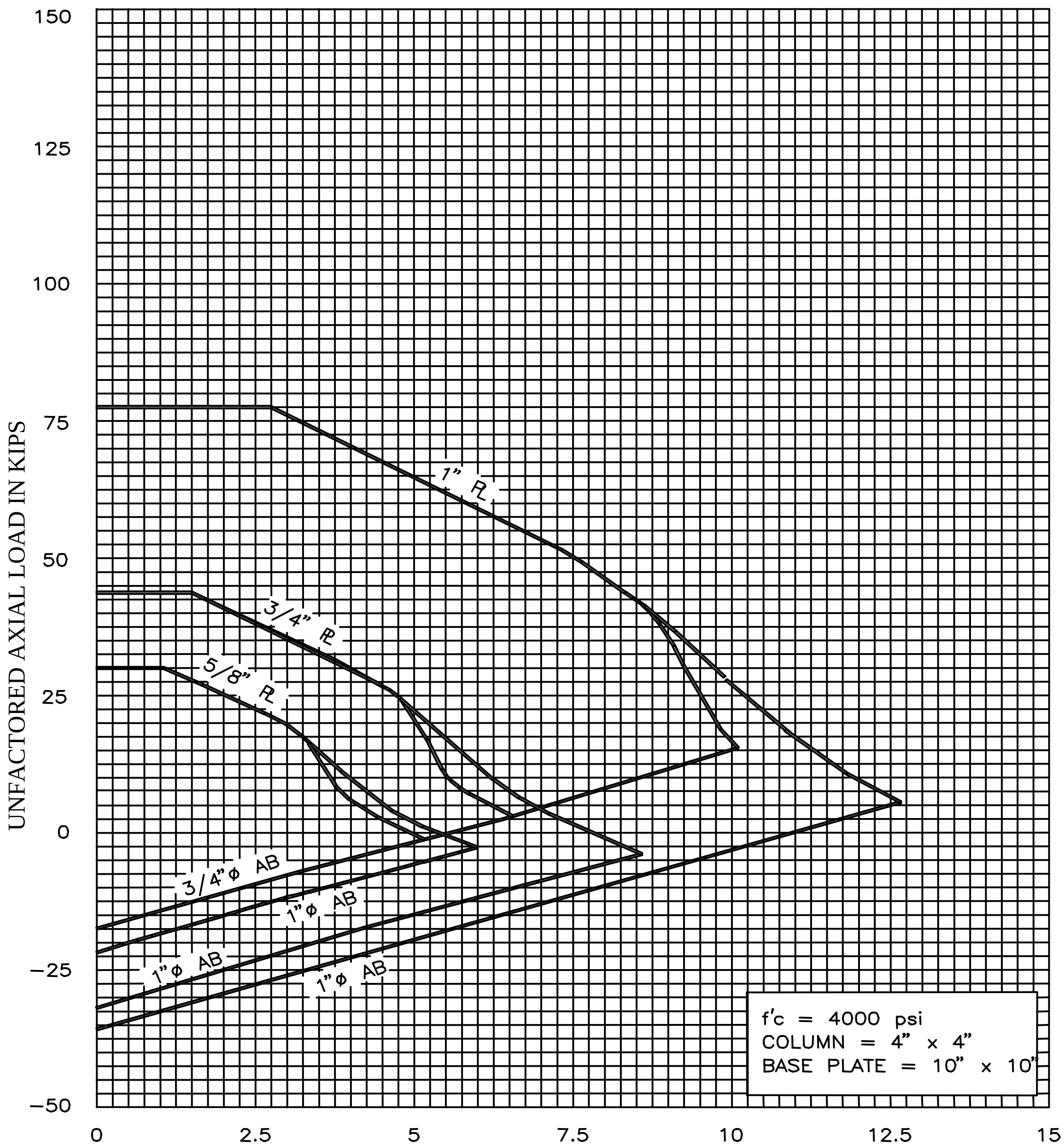
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves

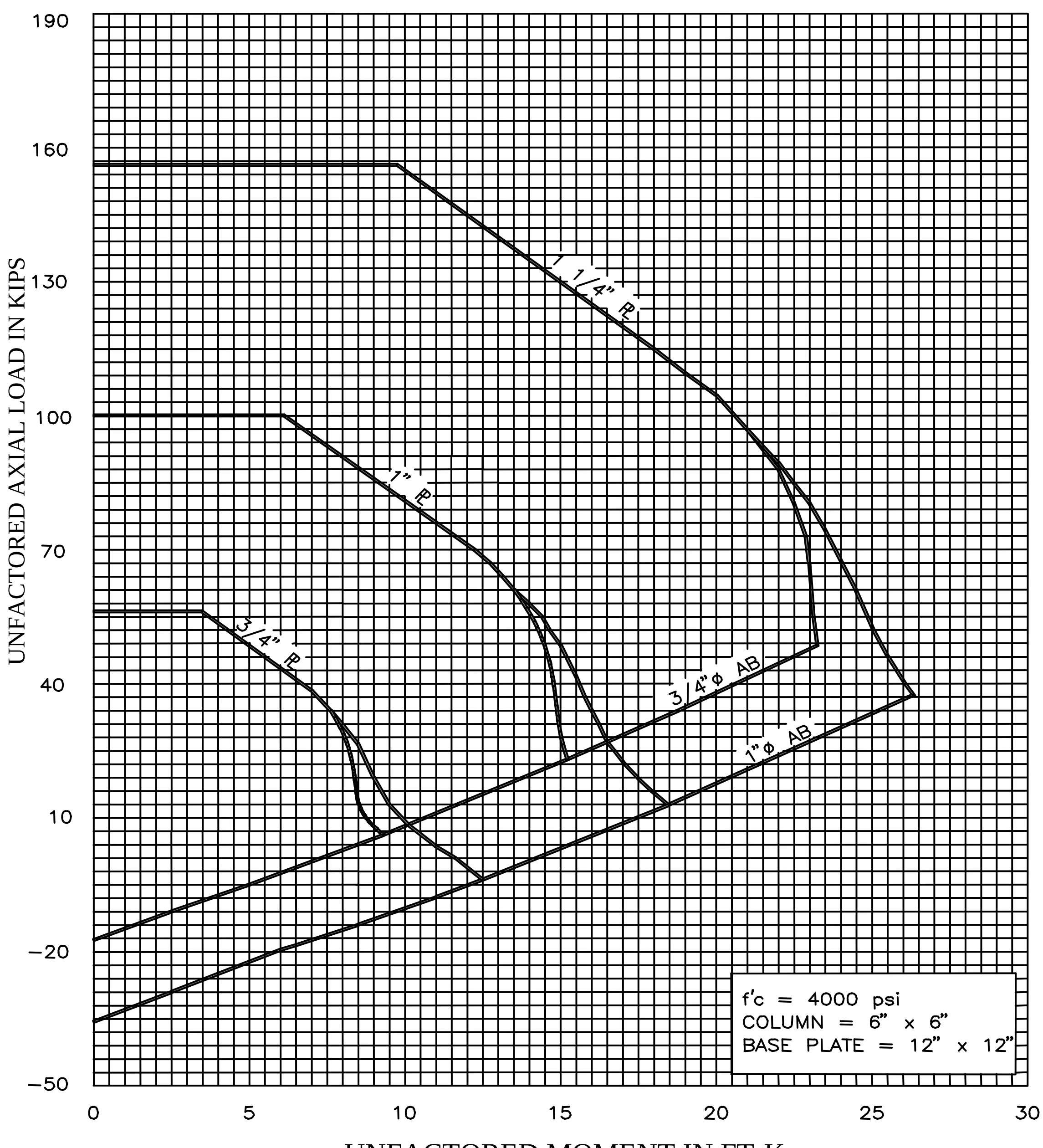


UNFACTORED MOMENT IN FT-K

Base Plate Capacity Curve 10

BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 11

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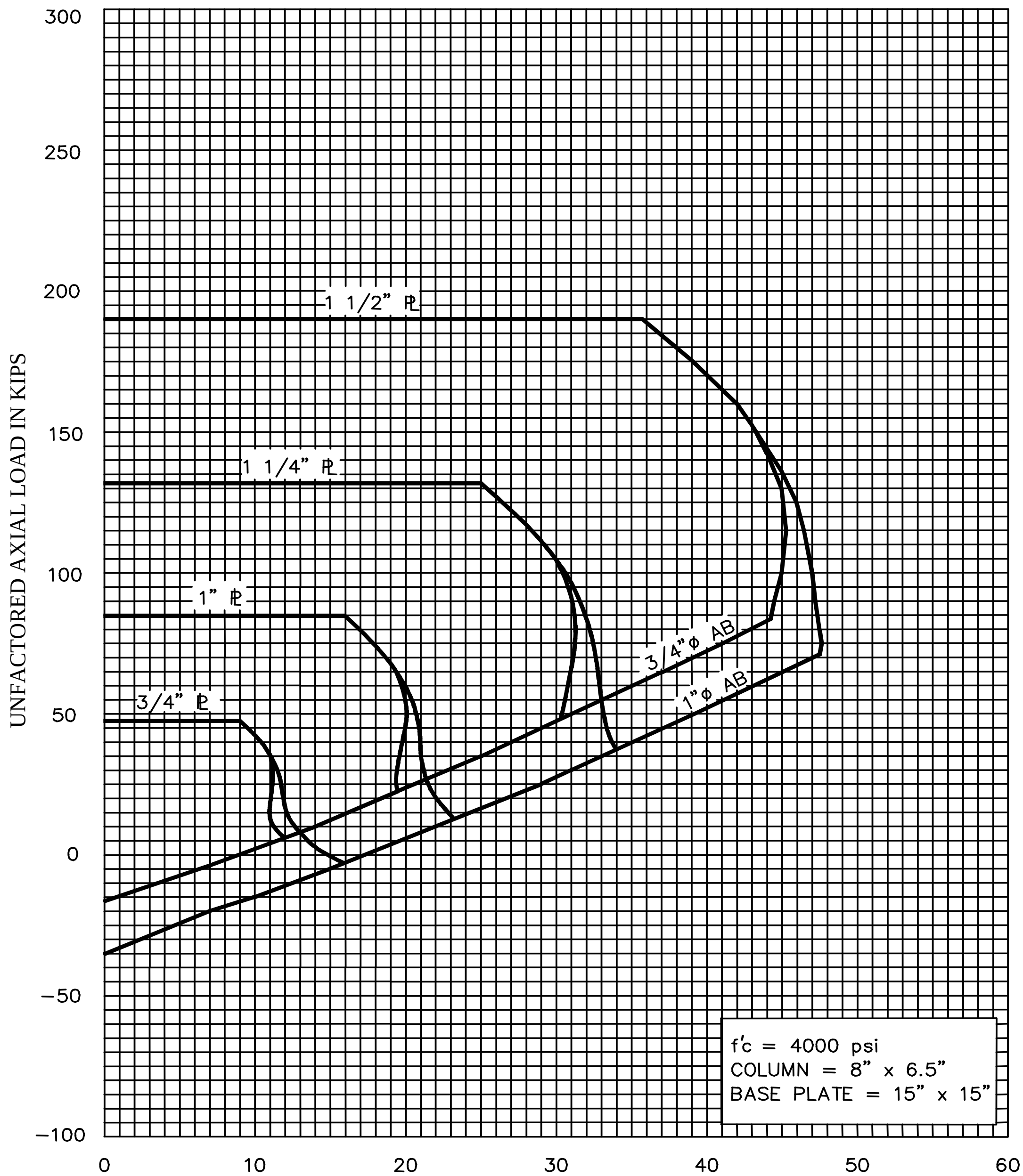
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



UNFACTORED MOMENT IN FT-K
Base Plate Capacity Curve 12

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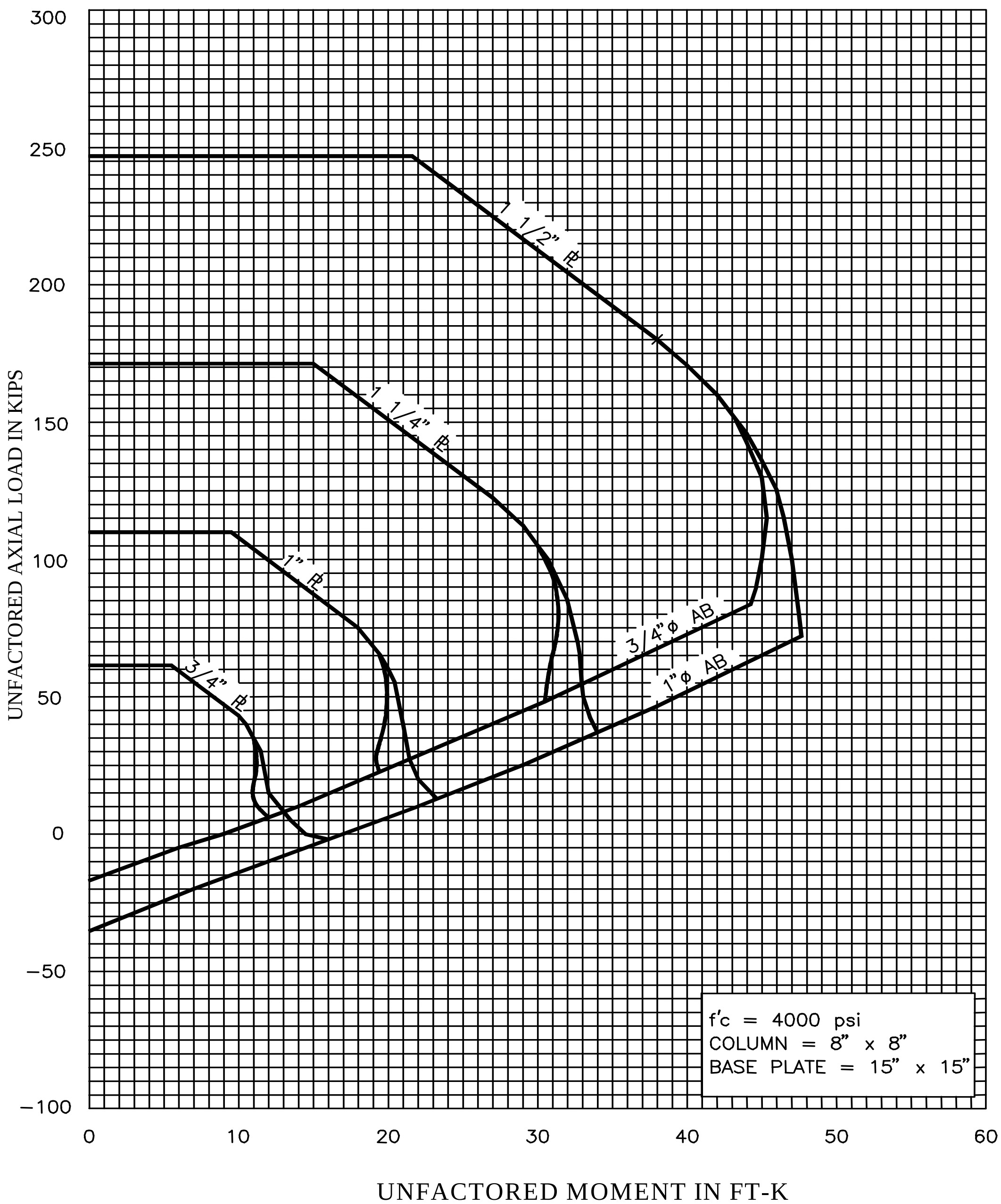
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 13

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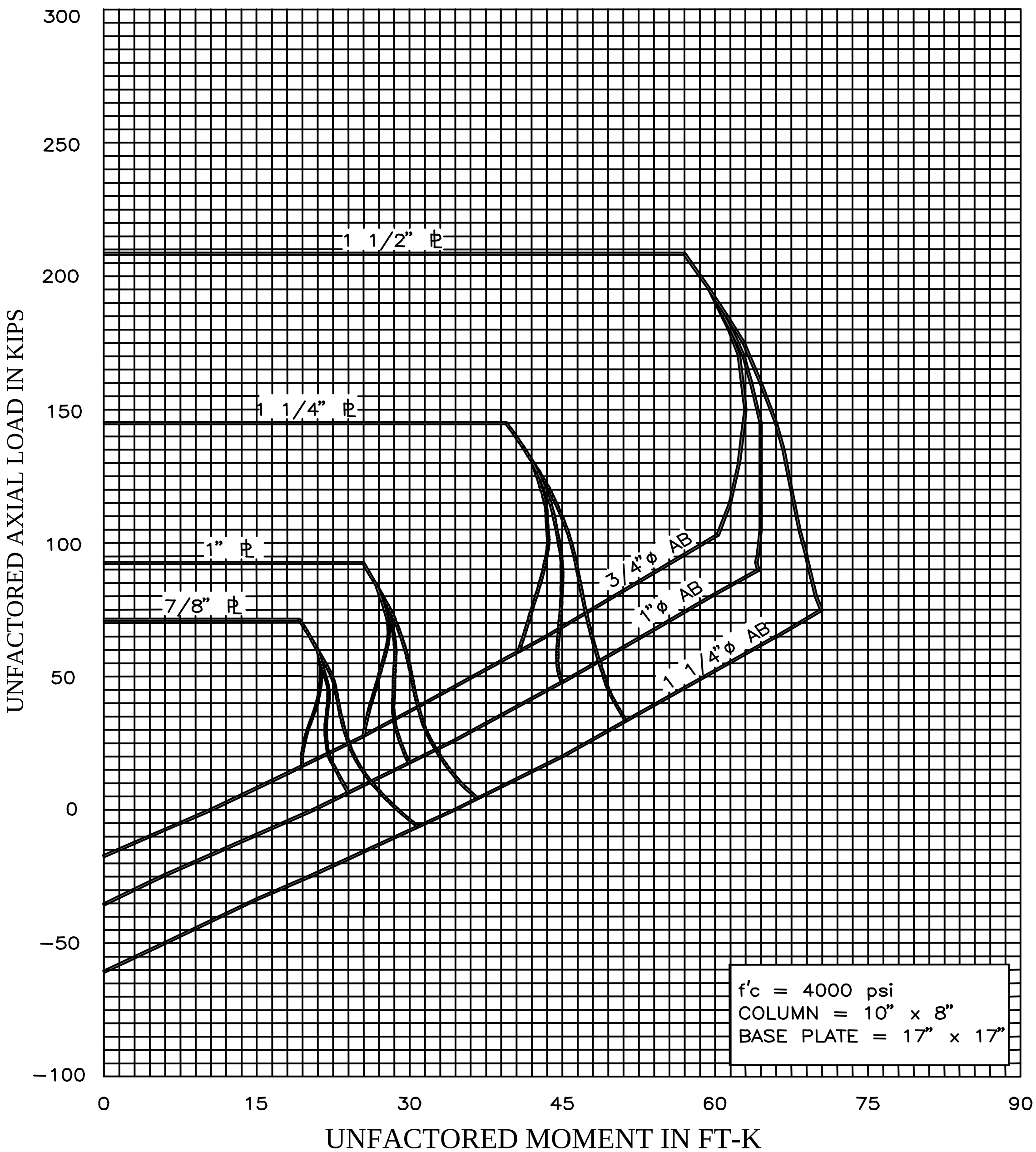
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 14

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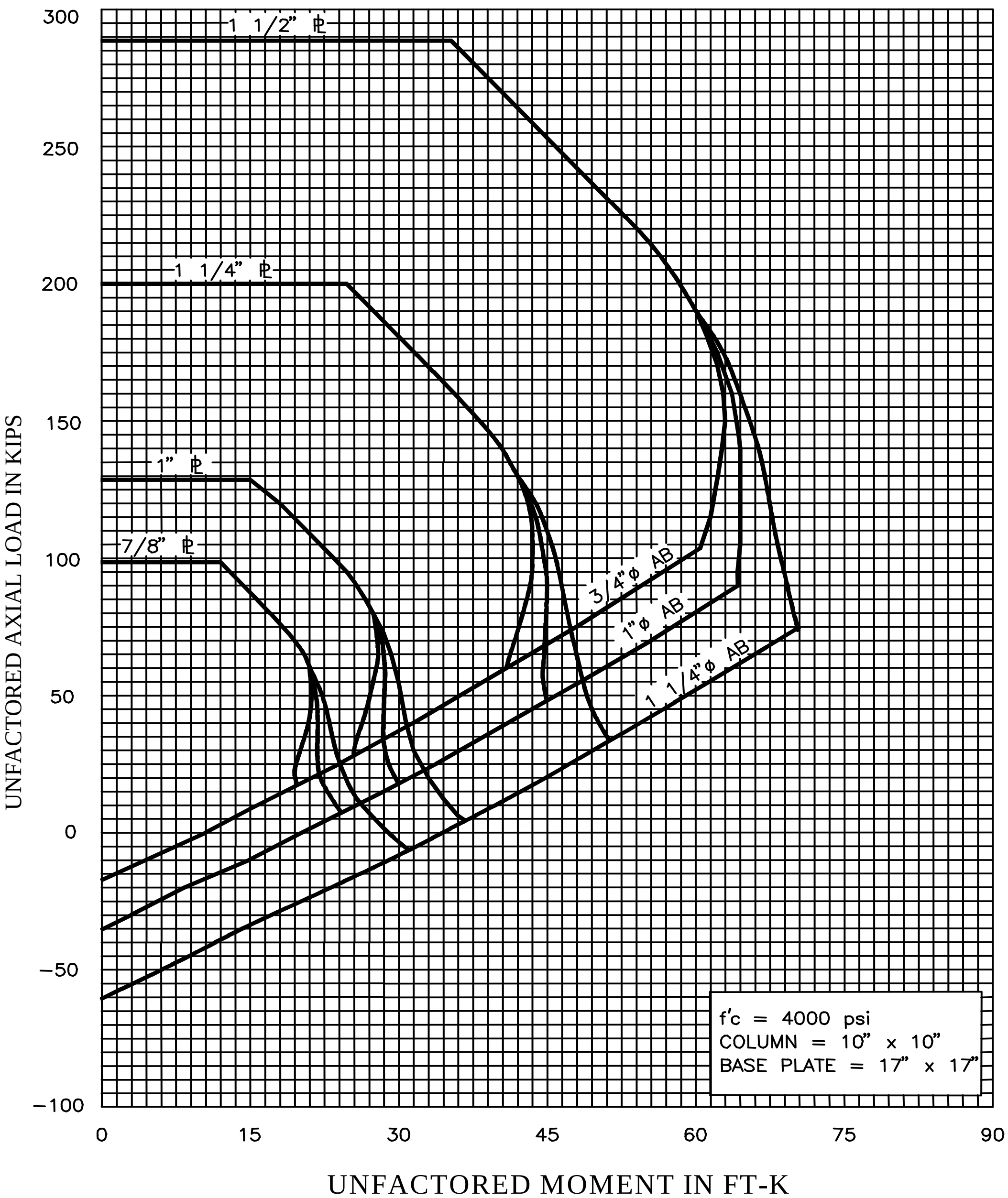
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 15

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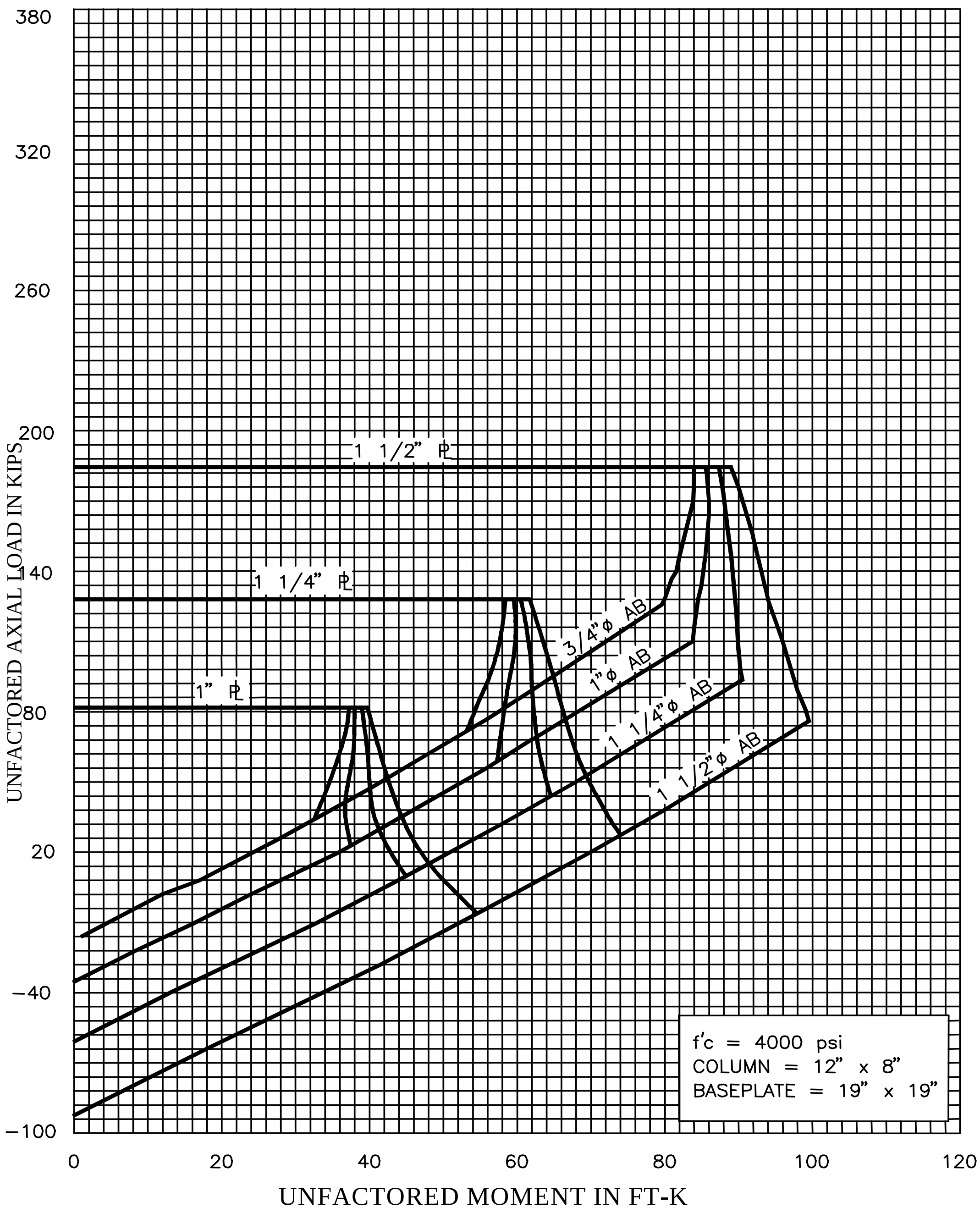
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 16

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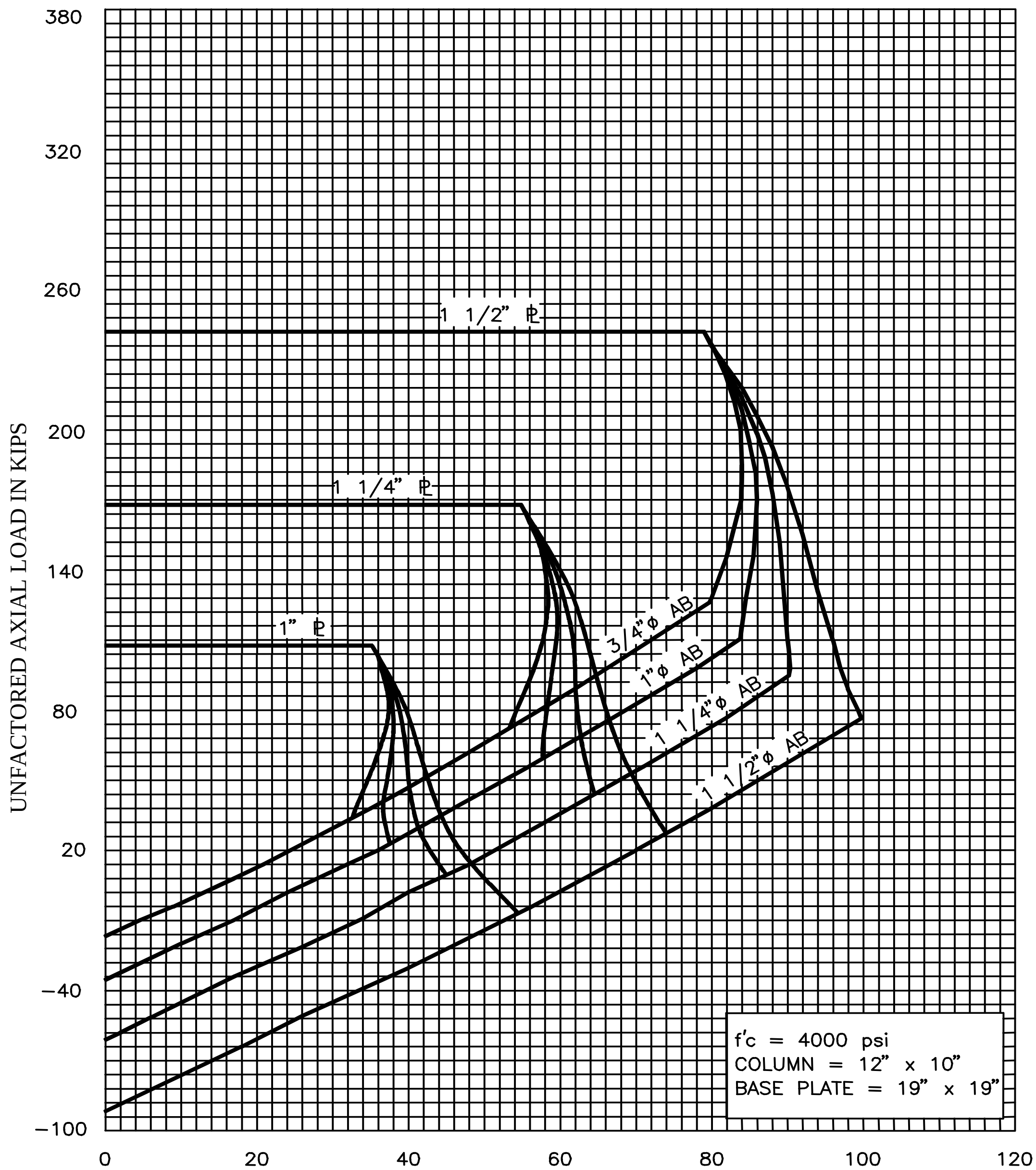
BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



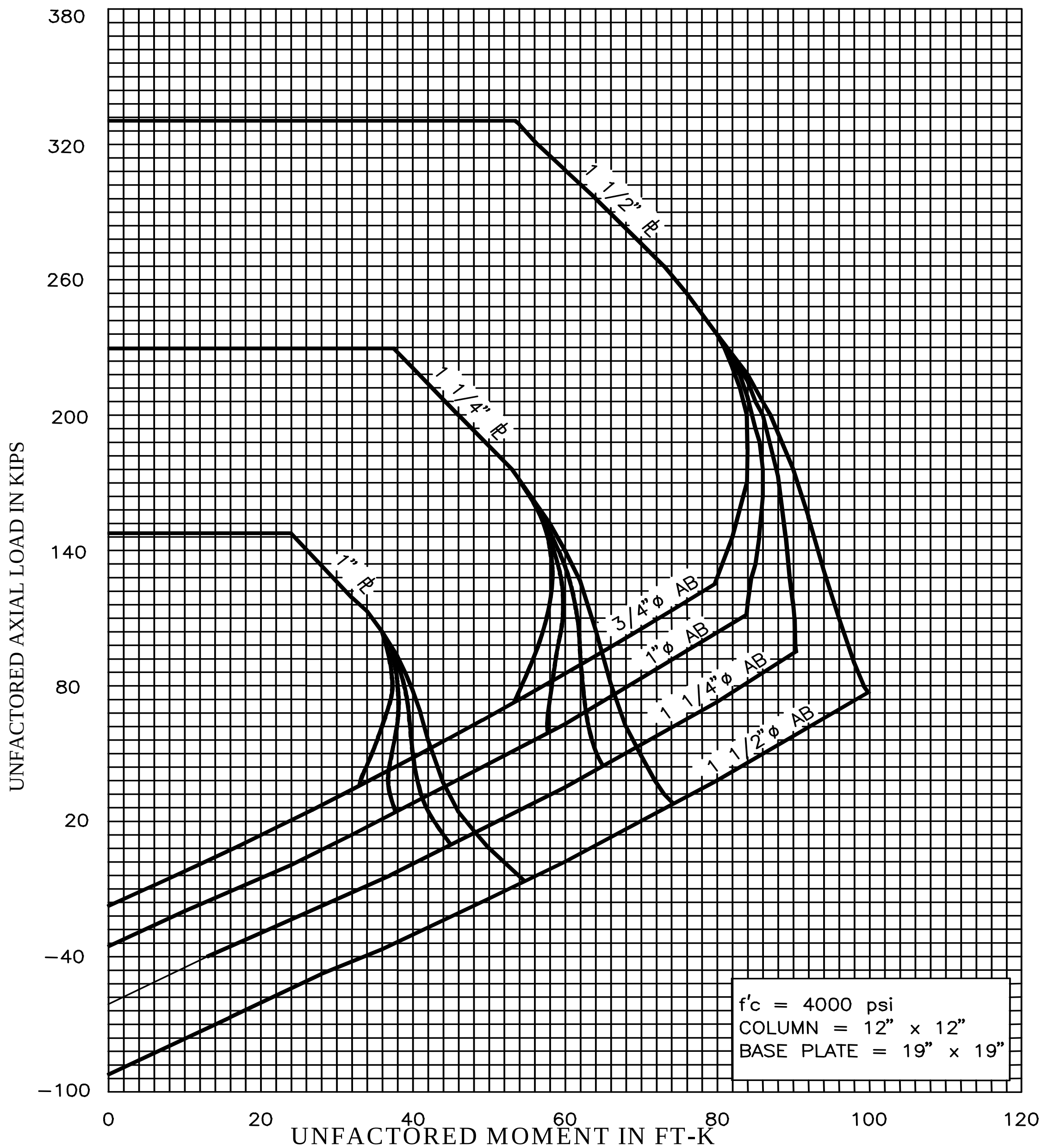
UNFACTORED MOMENT IN FT-K

Base Plate Capacity Curve 17

FLUOR

BASE PLATE DESIGN CRITERIA

Base Plate Capacity Curves



Base Plate Capacity Curve 18